

Realization-Dependent Microstructure and Statistical Morphology Emulation in Ballistic Helical GLAD

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(Dated: 20 August 2026)

Glancing-angle deposition (GLAD) grows structured thin films by depositing material onto a substrate held at an oblique angle; a single control parameter, the deposition angle α , governs both how porous the resulting film is and how differently it interacts with light polarized along different directions – properties relevant to gas sensors and polarization-based optical devices. What has not been established is how much of a film’s internal structure a simplified, physics-based computer model can actually predict from α alone. We address this with a ballistic simulator, in which atoms are treated as hard spheres that stick wherever they first touch the growing film, with no surface diffusion, across twelve helical-rotation Cu film configurations ($\alpha = 60^\circ\text{--}89^\circ$, height $h = 300\text{ nm}$, helical pitch = 150 nm). Void fraction (the empty-space fraction of the film) increases monotonically from 88.1% to 98.5% with α , and is highly reproducible: repeat runs with different random seeds agree to within 0.25% at this box/height convention (this reproducibility degrades at larger lateral box size; Appendix A). By contrast, we identify a clear limit to what is predictable: a corotating-frame oracle test shows that the exact position of an individual column carries no signal that generalizes across different random realizations of the same growth conditions – a model given only spatial coordinates as input cannot recover it – and we report this as a deliberate, central finding rather than a shortfall to be optimized away. What is predictable and reproducible is a statistical descriptor of the film’s internal structure, the two-point pair correlation $h_g(r; \alpha)$, which varies smoothly with α . A compact neural emulator (CORRNIF) reproduces angles held out of training with $R^2 = 0.992$, and a genuinely unseen prospective angle ($\alpha = 72^\circ$, later confirmed by a dedicated simulation) with $R^2 = 0.9907$ – but neither result is more accurate than simple linear or PCHIP (standard curve-fitting) interpolation on the same held-out angles ($R^2 = 0.9909$), so predictive accuracy alone is not an advantage over much simpler methods. CORRNIF’s one demonstrated practical advantage is different: gradient descent through the frozen, trained model recovers the deposition angle behind a held-out target curve from five different starting points to the same value, matching an independent grid search to within 0.002° . An illustrative optical calculation (treating the film’s columns as aligned tungsten cylinders, via the Maxwell-Garnett effective-medium approximation) predicts a large difference in refractive index for light polarized parallel versus perpendicular to the columns (form birefringence $\Delta n \sim 0.3\text{--}0.8$), but strong absorption for one polarization direction and high sensitivity to exactly how the solid fraction is defined mean this cannot be claimed as a usable polarization-device design window; we report it as a loss-aware screening illustration, not a device prediction. Together, these results define a simulation-only workflow for ranking candidate helical GLAD geometries by their statistical morphology, with the limits of predictive reach established as an explicit finding rather than left implicit, and with quantitative optical and material-specific calibration left to dedicated experiments.

I. INTRODUCTION

Glancing-angle deposition (GLAD) exploits extreme substrate inclination ($\alpha > 70^\circ$) to produce self-assembled nanocolumnar thin films with tunable porosity and helical geometry, an approach whose sculptured-column control originates with the chiral thin films of Robbie, Brett, and Lakhtakia¹ and has since been reviewed extensively^{2–6}. The physical origin and evolution of the columnar, shadowing-driven morphology itself has been

characterised in detail^{7,8}. The helical architecture generates strong geometric form birefringence: the ordinary and extraordinary refractive indices of the nanostructured film differ because oriented columnar inclusions interact differently with polarisation states parallel and perpendicular to the helix axis⁹. This tuneable optical anisotropy underlies the application of GLAD films in ellipsometric waveplates, chiral-selective filters, and label-free polarimetric biosensors where large phase retardation between the two polarisation eigenstates is required^{10,11}. Recent demonstrations of Cu_2O GLAD films as gas-sensing platforms underline the practical urgency of predictive design tools: surface roughness and optical anisotropy both depend sensitively on α , and

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model-guided angle selection is critical for optimising sensor response¹². Realising this design potential demands models that reliably link α to the structural filling fraction $f(\alpha)$ and thence to effective optical anisotropy.

Ballistic simulation models, which track straight-line atom trajectories without thermally activated surface diffusion, are the fastest tool for screening deposition-angle effects on void fraction and columnar geometry at the nanometre scale¹³. The self-shadowing tangent rule underlying such models traces back to the microfractographic columnar-tilt observations of Nieuwenhuizen and Haanstra¹⁴, was formalised into predictive columnar-growth models by Tait, Smy, and Brett¹⁵ and Lichter and Chen¹⁶, and was extended to full three-dimensional ballistic simulation—of which the present simulator is a direct descendant—by Smy *et al.*¹⁷; the effective surface area generated by such oblique shadowing growth has been analysed numerically by Suzuki and Taga¹⁸. A more recent, independently implemented ray-traced Monte-Carlo 3D GLAD simulator, validated against the same column-growth-angle relation, is given by Kulkarni *et al.*¹⁹, illustrating that this modelling family continues to be actively re-implemented rather than being a single inherited codebase. Despite their computational efficiency, ballistic models are known to over-predict porosity relative to experimentally deposited films because adatom surface diffusion tends to densify the structure after impact^{20,21}. The recent success of neural implicit fields (NIFs)—compact multi-layer perceptrons that map continuous coordinates to scalar field values, as established for shape and scene representation by Sitzmann *et al.*²², Park *et al.*²³, and Mescheder *et al.*²⁴, and made practical at high spatial resolution via multiresolution hash encodings²⁵—in representing three-dimensional structures has raised the question of whether such models can learn and interpolate GLAD column morphology directly from simulation data, enabling rapid parameter-space exploration without re-running the simulator at every deposition angle²⁶. However, neither the fundamental predictability of per-voxel column positions nor the reproducibility and interpolability of statistical descriptors extracted from GLAD simulation data has been systematically established.

Here we address this gap with a combined simulation-characterisation and machine-learning study of the bead-sphere ballistic GLAD simulator across twelve helical Cu configurations spanning $\alpha = 60^\circ$ – 89° ($h = 300$ nm, pitch = 150 nm). Our central machine-learning contribution is a principled two-level investigation of data-driven morphology modelling. First, a corotating-frame oracle experiment shows empirically that absolute per-voxel column positions are not a target a coordinate-conditioned neural implicit field can recover under the multi-alpha stochastic-nucleation protocol of this simulator: the best fixed spatial template gives no gain over the bulk-filling-fraction baseline, because column positions are re-randomised in every independent run. Second, we demonstrate that the normalised two-point

pair correlation $h_g(r; \alpha)$ —an approximately reproducible ensemble descriptor under the tested seeds—is instead smoothly interpolable: a compact parametric emulator (CORRNIF) trained on six deposition angles reproduces $h_g(r; \alpha)$ at unseen interior angles with mean leave-one-angle-out $R^2 = 0.992$. A prospective validation at $\alpha = 72^\circ$, genuinely absent from the training set and subsequently confirmed by a dedicated simulation, is reproduced by CORRNIF with $R^2 = 0.9907$, while a simple linear interpolation gives $R^2 = 0.9909$ on the same corrected target. We additionally quantify the cross-material, cross-method discrepancy between the ballistic model and published optical void fractions and, combining the extracted filling fractions $f(\alpha)$ with the aligned-cylinder Maxwell-Garnett effective-medium model²⁷, provide a loss-aware birefringence screening illustration for high-index columnar geometries.

Concretely, this paper answers four questions that an experimentalist choosing a deposition angle, or a modeller deciding what to simulate next, would need answered before trusting a ballistic GLAD simulation:

- Q1** How does bead-sphere void fraction depend on α in this ballistic model, how reproducible is it across independent stochastic runs, and how does it compare quantitatively with published optical void fractions of other GLAD metals?
- Q2** Does the film’s internal structure vary with depth in a way that matters for property predictions, and can a small set of bulk morphology descriptors (openness, anisotropy, width gradient) be summarised in a low-dimensional space across the angle sweep?
- Q3** Can a machine-learning model (a neural implicit field) learn to predict GLAD morphology from simulation data, and if so, at what level—individual column positions, or ensemble statistics—is such a prediction actually possible and reproducible?
- Q4** What do the resulting morphology descriptors imply, even only illustratively, for the optical anisotropy that end applications (polarimetric sensors, waveplates) depend on?

The Conclusion (Section VI) closes against these four questions in the same order. Specifically, we report: (i) the bead-sphere void-fraction trend and its three-seed reproducibility (Section III A, Q1); (ii) depth-resolved void-fraction profiles that reveal growth-front evolution (Section III B, Q2); (iii) a morphology-space analysis by principal component analysis (Section III C, Q2); (iv) a quantitative cross-material, cross-method discrepancy of +20.7 pp against published optical void fractions (Section III D, Q1); and (v) a full analysis: a corotating-frame oracle bounding absolute-position recovery, CORRNIF pair-correlation interpolation with interior leave-one-angle-out $R^2 = 0.992$, prospective validation at $\alpha = 72^\circ$ (CORRNIF $R^2 = 0.9907$), and an illustrative

loss-aware Maxwell-Garnett birefringence screening (Section IV, Q3 and Q4). Section V discusses implications for data-driven GLAD modelling and the limits of effective-medium optical screening.

II. COMPUTATIONAL METHODS

A. Simulation framework

Figure 1 summarises the mechanism before the formal description below: an experimentalist setting the deposition angle α on a GLAD chamber is directly setting the tilt of the incoming atom beam relative to the substrate normal, and the programmed substrate rotation is what turns this into a helix rather than a simple tilted column.

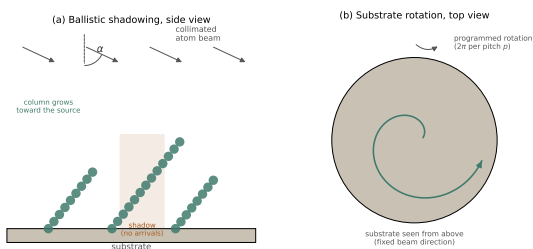


FIG. 1. The two knobs an experimentalist turns on a GLAD chamber – beam tilt α and substrate rotation – are exactly the two inputs to this simulator; everything else (columns, shadowing, voids) is an emergent consequence of these two settings plus first-contact sticking. Schematic of the ballistic deposition mechanism (conceptual diagram, not simulation output; compare Figure 2 for the simulator’s actual output). (a) Side view: atoms arrive as a collimated beam at angle α from the substrate normal and stick irreversibly on first contact, with no subsequent surface diffusion. A column that nucleates first casts a geometric shadow (shaded region) that suppresses further arrivals immediately behind it – this is the entire origin of the void fraction trend in Section III A. (b) Top view: the substrate rotates continuously during growth, tracing one full 2π turn per helical pitch p , which is what makes the resulting columns helical rather than straight.

Physically, the simulator asks a simple question: if atoms travel in straight lines from the source and stop irreversibly the instant they touch another atom or the substrate—i.e. if the only thing shaping the film is geometric shadowing, with no atom motion after landing—how much of the deposited volume ends up empty (void) rather than filled (solid), and how does that depend on the deposition angle α that an experimentalist sets on the chamber? Simulations were performed with a custom bead-sphere ballistic deposition code. Atoms are launched from a collimated beam incident at angle α from the substrate normal. Each atom follows a straight-line (ballistic) trajectory and sticks irreversibly to the first atom it contacts; no surface diffusion, relaxation, or evaporation is included. The film grows on a square substrate of lateral size $L_x \times L_y = 100 \text{ nm} \times 100 \text{ nm}$

with periodic (minimum-image) boundary conditions in the two in-plane directions. The model contains no material-specific physics: “Cu” enters only through the hard-sphere atomic radius (i.e. a Cu-scale length), not through any Cu-specific diffusion, sticking, or energetic parameter. The simulator is therefore a *generic hard-sphere ballistic helical GLAD model parameterised at a Cu atomic scale*, and the geometric outputs apply to any material only insofar as its real morphology is shadowing-dominated; the later substitution of tungsten optical constants (Section IV E) is an illustrative material substitution, not a calibrated W-film model.

B. Helical substrate and parameter space

Physically, this subsection reproduces in simulation the two knobs an experimentalist actually turns during a helical GLAD run: the fixed deposition angle α (substrate tilt relative to the vapour beam) and the programmed substrate rotation that traces out one helical turn per pitch length p ; the goal is to isolate how much of the resulting morphology change is due to α alone, holding the rotation program and target film height fixed. A helical hard-sphere nanostructure template is defined with target height $h = 300 \text{ nm}$ and helical pitch $p = 150 \text{ nm}$ (two full turns within the film). The pitch is a controlled numerical reference derived from the standard helix pitch relation using the vertical-growth-rate convention; it is used here to isolate deposition-angle dependence and is not presented as an experimentally calibrated Cu pitch. Twelve configurations are simulated for the main morphology and reproducibility study: deposition angles $\alpha \in \{60, 65, 70, 75, 80, 80, 80, 85, 85, 85, 87, 89\}^\circ$, where three configurations at $\alpha = 80^\circ$ and three at $\alpha = 85^\circ$ use independently seeded random-number generators to assess stochastic reproducibility. A thirteenth simulation at $\alpha = 72^\circ$ was subsequently performed exclusively as a prospective validation case for the CORRNIIF interpolation (Section IV C); it was excluded from CorrNIF training and from all main-dataset descriptive analyses. The deposition angle α is measured from the substrate normal, and the helix is generated by programmed azimuthal rotation of the substrate during growth (one 2π azimuthal turn per pitch length p). Table I collects the simulation and analysis parameters for reproducibility; quantities not separately parameterised in this ballistic model (beam angular divergence, arrival-position distribution beyond uniform random azimuth, explicit nucleation/sticking rules) are defined by the simulation code (available from the corresponding author) rather than by tunable inputs.

Before the quantitative characterisation that follows, Figure 2 shows directly what the simulator produces: real voxelised cross-sections of a simulated film, not a schematic. At $\alpha = 89^\circ$, the lateral (xy) slice shows isolated, well-separated columns rather than a continuous film, and the vertical (xz) slice shows these columns

as tilted streaks – the geometric shadowing signature that the tangent rule and the void-fraction trend in Section III A both quantify. At $\alpha = 60^\circ$, the same lateral slice is nearly fully covered, consistent with the much lower void fraction at this shallower angle.

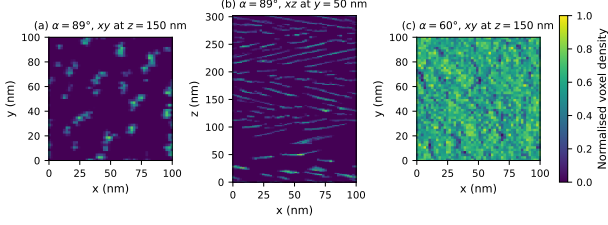


FIG. 2. Direct visual evidence of the shadowing mechanism that the rest of this paper characterises quantitatively: isolated, tilted columns at high α versus a near-continuous film at low α . Voxelised density cross-sections of simulated Cu GLAD films ($h = 300$ nm, pitch = 150 nm, seed 1, 2 nm voxels). (a) Lateral (xy) slice at mid-height ($z = 150$ nm), $\alpha = 89^\circ$: isolated columns, 8.2% occupied. (b) Vertical (xz) slice at $y = 50$ nm, $\alpha = 89^\circ$: the same columns appear as tilted streaks, the direct signature of ballistic shadowing along the deposition direction. (c) Lateral slice at $\alpha = 60^\circ$ for comparison: 83.5% occupied, a near-continuous film at this shallower angle. Despite occupying only 8.2% of the volume at $\alpha = 89^\circ$, the column network remains fully percolated (largest connected fraction = 0.974, i.e. essentially all deposited material belongs to one connected structure spanning the film).

C. Void-fraction definition

Physically, this quantity is the porosity-like measure of the film: the fraction of the deposited volume that is empty space between column atoms rather than solid material, computed geometrically from the known hard-sphere size and count of the deposited atoms rather than from an imaging or scattering measurement. The bead-sphere void fraction is defined as:

$$P = 1 - \frac{NV_b}{L_x L_y z_{\max}}, \quad V_b = \frac{4}{3}\pi R^3, \quad (1)$$

where N is the number of deposited atoms, R is the atom radius, and $z_{\max}$ is the height of the topmost atom. This quantity directly corresponds to the open volume in the idealised hard-sphere packing and is logged by the simulator at the end of each simulation run. It differs from the voxel-mean quantity $1 - \langle \rho_{\text{voxel}} \rangle$ used for density-field analyses (Sections III B and III C).

D. Voxel density field

Physically, this step turns the raw list of atom positions into something directly comparable to a depth-resolved

TABLE I. Simulation and analysis parameters. Quantities marked “code” are fixed by the ballistic simulator code rather than exposed as tunable inputs.

Parameter	Value	Source/status
<i>Geometry and domain</i>		
Substrate lateral size $L_x \times L_y$	100×100 nm	prescribed in this work
Lateral boundary conditions	periodic (minimum-image), XY	simulator setting
Target film height h	300 nm	prescribed in this work
Final height range (12 runs)	300.0 nm–300.1 nm	simulation result, this work controlled
Helical pitch p (turns)	150 nm (2 turns in h)	numerical reference; not Cu-calibrated
Deposition-angle convention	polar, from substrate normal	simulator convention derived from
Azimuthal rotation law	programmed, 2π per pitch length	prescribed pitch
<i>Ballistic deposition model</i>		
Hard-sphere radius R (diameter $2R$)	0.1 nm (0.2 nm)	Cu-scale model parameter (optical screening subsection uses the recalibrated $R = 0.128$ nm campaign; see Section IV E)
First-contact criterion	$2R$ physical contact	simulator rule
Sticking probability	1 (irreversible, first contact)	simulator rule
Surface diffusion / re-emission / evap.	none / none / none	prescribed model scope
Re-emission, angular divergence	none; collimated beam	prescribed model scope
XY launch distribution	uniform random over the cell	simulator rule
Initial seed layer	10^4 seed atoms	numerical setting, this work
Nominal deposition rate	0.2 nm/s (dense-equiv.)	numerical setting, this work
Substrate temperature	not active in the ballistic physics hard-sphere	not used by this model
“Cu” material content	radius (a Cu-scale length) only	not a calibrated Cu model
Deposited atoms per run	1.1×10^7 – 1.6×10^8	simulation result, this work
Stopping criterion	target height $h = 300$ nm reached	prescribed in this work
Random seeds (RNG)	1 per angle; 3 at $80^\circ, 85^\circ$	numerical setting, this work
<i>Voxelisation and correlation</i>		
Voxel grid	$50 \times 50 \times 151$ (2 nm XY, 2 nm Z)	numerical setting, this work
Voxelisation kernel	Gaussian bead-centre histogram	analysis setting
Kernel / density normalisation	max-normalised, $\rho_{\text{voxel}} \in [0, 1]$	analysis setting
Occupancy	$\rho_{\text{voxel}} \in [0, 1]$	analysis

microstructure map, analogous to a reconstructed FIB-SEM or nano-CT tomogram of a real film, so that how dense or open the structure is at a given depth and lateral position can be tracked separately from the single, film-averaged void fraction of Section II C. For depth-resolved and PCA analyses, atom positions are projected onto a $50 \times 50 \times 151$ voxel grid (2 nm XY resolution, 2 nm Z resolution, spanning the 100 nm $\times$ 100 nm $\times$ 300 nm cell) using a Gaussian bead-centre kernel. The resulting normalised density field $\rho_{\text{voxel}}(\mathbf{r}) \in [0, 1]$ encodes the local atom-sphere overlap fraction, with the continuous three-dimensional material distribution sampled on a regular spatial grid as in the tomographic analogy above. The Level-A run at $\alpha = 60^\circ$ had a corrupted final checkpoint (active-slab resume duplication artefact, physically impossible atom count of 163 M vs. the expected ~ 85 M at 300 nm); both its bead-sphere value ($P = 77.2\%$) and its voxel are invalid and excluded from all analyses. The openness-trend curve uses $P = 88.1\%$ from a clean single-shot replacement run (literature-matched benchmark, independent simulation from a validated starting state; SHA-256 c7509b21...).

E. Morphological descriptors and PCA

Physically, a single voxel field contains far more information than an experimentalist can compare across many deposition angles at a glance; the goal here is to compress each film down to a handful of numbers that capture how open it is, how directionally lopsided its columns are, and whether the columns broaden or narrow with depth, and then to check whether deposition angle alone organises these numbers into a simple, low-dimensional trend. For each of the eleven valid voxel files we use three non-redundant descriptors: (i) openness proxy $1 - \langle \rho_{\text{voxel}} \rangle$; (ii) XY anisotropy ratio (ratio of power along orthogonal in-plane axes); (iii) apparent width-gradient proxy (linear fit to column-width growth with depth; used only as a descriptive feature, not interpreted as a physical fanning exponent). We deliberately exclude the mean filling fraction $\langle \rho_{\text{voxel}} \rangle$, which is exactly complementary to the openness proxy (openness = $1 - \langle \rho_{\text{voxel}} \rangle$): including both perfectly collinear measures double-counts the density axis and artificially inflates the variance attributed to it on the leading component. Features are standardised before PCA; principal components are computed via SVD, and variance fractions and loadings are reported. The eleven observations include repeated seeds at 80° and 85° , which we keep as individual points (rather than averaging) so that the seed-to-seed scatter is visible. The result is therefore an exploratory descriptor-space summary, not a demonstration of a general morphology manifold.

F. Statistical morphology pipeline

Corotating-frame pair correlation

Physically, this descriptor asks how likely a column is to be found at a given lateral distance from another column, once the steady rotation of the substrate has been numerically “undone” so that a static column arrangement, rather than a rotating helix, is being compared from film to film; it is the real-space analogue of the column-spacing information that a scattering experiment (SAXS/GISAXS) would extract from the diffraction pattern of the same film. The corotating-frame correlation starts from the binary occupancy field $I(\mathbf{r}) = [\rho_{\text{voxel}} > 0.5]$, from which the filling fraction $f = \langle I \rangle$ is taken. Each z -slice $I(\cdot, \cdot, k)$ is then derotated by the local helix phase $\phi(k) = 2\pi k/p_{\text{vox}}$, where $p_{\text{vox}} = p/\Delta z$ is the pitch in voxels, using bilinear interpolation to avoid the pixel-level jitter that nearest-neighbour resampling accumulates over the 151 z -slices. Because the domain is periodic, the bilinear resampling uses periodic (wrap) source coordinates rather than edge-clamped coordinates; the clamped variant violates per-slice mass conservation by up to 10% near the boundary and is therefore replaced here. Switching from the edge-clamped to the periodic-wrap derotation perturbs h_g by a mean absolute difference of 0.007 over the full curve (0.015 at short range, where the steep near-origin drop amplifies small horizontal shifts; maximum pointwise 0.035). This is comparable to the intrinsic seed-to-seed scatter (≈ 0.008 , below) and leaves the angle ordering and every qualitative conclusion unchanged (validation tests included with the analysis code). Bilinear derotation maps the binary slice to a *continuous* occupancy field $J \in [0, 1]$, for which the binary identity $\langle J^2 \rangle = \langle J \rangle$ no longer holds; we therefore normalise by the empirical zero-lag (continuous-covariance) variance rather than by the binary value $1/f$ (Eqs. 3–4). The circular autocorrelation $C(r) = \langle J(\mathbf{x})J(\mathbf{x}+r) \rangle$ is obtained by FFT, radially averaged, and the result is averaged over all z -slices to yield the ensemble-mean corotating-frame curve $\langle h_g(r; \alpha) \rangle_z$. This is computed for the seven single-seed configurations $\alpha \in \{65, 70, 75, 80, 85, 87, 89\}^\circ$. The lateral domain is 100 nm with periodic boundaries, and the radial average is taken to $r_{\text{max}} \approx 24$ voxels (≈ 48 nm), i.e. close to half the box width $L/2$. Under the periodic (minimum-image) convention the correlation near $r \rightarrow L/2$ is constrained by the finite domain, so the curves are informative mainly at short range ($r \lesssim 10$ voxels ≈ 20 nm, the regime that dominates the descriptor and the emulator targets) and the long-range tail should be read with caution. A box-size convergence study (e.g. 100, 150, 200 nm at representative angles) would be required to confirm that the short-range correlation is free of finite-size or periodic artefacts; this requires new simulations and is left as future work. (A related but distinct check – bulk atom-density convergence with box size at fixed pitch, not the correlation function itself – was performed

and found NOT converged at $L = 100$ nm; see the box-size limitation in Section V D.)

CorrNIF architecture and training

CORRNIF is a compact parametric emulator that maps deposition angle α and radial distance r to the predicted pair-correlation value $h_g(r; \alpha)$ at any combination of these two inputs—analogous to a parametric dispersion model that evaluates a refractive-index spectrum at any wavelength and filling fraction without re-running a full electromagnetic simulation.

A compact neural implicit field (CORRNIF) maps $(\alpha_{\text{norm}}, r_{\text{norm}}) \rightarrow h_g(r; \alpha)$, where $\alpha_{\text{norm}} = (\alpha - 74.5^\circ)/14.5^\circ$ and $r_{\text{norm}} = r/r_{\text{max}}$. The radial coordinate is encoded via $n_{\text{freq}} = 12$ random Fourier features²⁶: $\mathbf{h}_0 = [\sin(2\pi\mathbf{B}r), \cos(2\pi\mathbf{B}r)]$, $\mathbf{B} \sim \mathcal{N}(0, \sigma_r^2)$, $\sigma_r = 2$. Three fully connected hidden layers (width 64, SiLU activation) receive angle conditioning (feature-wise linear modulation, FiLM) from α : $\mathbf{h}_{i+1} = \text{SiLU}[(1 + 0.1\gamma_i)\mathbf{W}_i\mathbf{h}_i + 0.1\beta_i]$, where (γ_i, β_i) are produced by an α -conditioned two-layer MLP (width 32, ReLU). Figure 3 summarises this architecture.

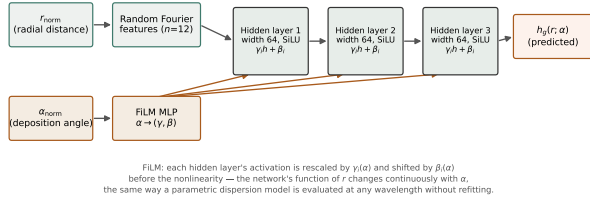


FIG. 3. The angle conditioning does not add a third input alongside r ; it reshapes the network's response to r at every layer, which is why a single trained model can be evaluated smoothly at angles it never saw during training (Section IV C). Schematic of the CORRNIF architecture (conceptual diagram). The radial coordinate r is Fourier-encoded and passed through three FiLM-conditioned hidden layers; the deposition angle α does not enter as a third raw input but instead rescales (γ_i) and shifts (β_i) each layer's activation via a small side network, analogous to how a parametric dispersion model is evaluated at any wavelength without being refitted at that wavelength.

Training uses the Adam optimiser (learning rate 10^{-3} , 1000 epochs, batch size 512, MSE loss on h_g). For the corrected analysis reported here, CORRNIF was re-trained from scratch on the periodic-wrap, continuous-covariance target; ten independent neural initialisations were run for each validation fold and for the prospective 72° case. Generalisation is evaluated by a seven-fold leave-one-angle-out (LOOA) protocol: in each fold, CORRNIF is trained on six deposition angles and used to predict $h_g(r; \alpha_{\text{held}})$ without any adaptation on the held-out angle—analogous to fitting an optical dispersion model to spectra measured at six incidence angles and evaluat-

ing its prediction at a seventh, held-out angle.

Maxwell-Garnett effective-medium model

Physically, this step asks what optical anisotropy a film with the simulated column filling fraction $f(\alpha)$ would show *if* its electromagnetic response could be approximated by smoothly averaging the column and void dielectric properties together, rather than by resolving the actual column geometry wave-optically—a standard first estimate of form birefringence, used here as a scoping tool rather than a precision prediction. Effective optical properties are derived from the Maxwell-Garnett (MG) effective-medium approximation for *aligned* cylindrical inclusions in a host medium²⁷ (Figure 4), a special case of the broader family of mixing formulas surveyed by Sihvola²⁹ and related to the symmetric effective-medium approximation of Bruggeman³⁰;

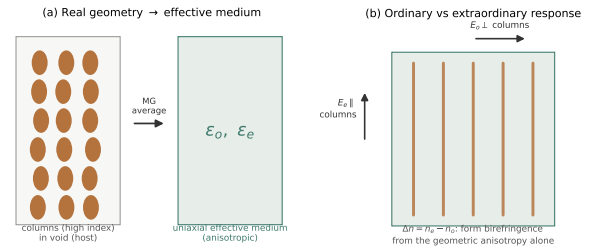


FIG. 4. The birefringence reported below is not an extra material property being added to the model; it falls directly out of averaging an already-anisotropic column geometry into a single effective medium. Conceptual illustration of the Maxwell-Garnett effective-medium approximation (not simulation output). (a) Aligned high-index cylindrical inclusions (columns) embedded in a low-index host (void) are replaced by a single uniaxial effective medium with two principal dielectric constants, ϵ_o and ϵ_e . (b) The two principal optical axes: the extraordinary axis E_e lies parallel to the columns, the ordinary axis E_o perpendicular to them. Form birefringence $\Delta n = n_e - n_o$ arises purely from this geometric anisotropy, before any intrinsic material birefringence is considered.

empirical extensions of this approach specifically fitted to the measured principal refractive indices of obliquely deposited oxide sculptured thin films (Ta_2O_5 , TiO_2 , ZrO_2) are given by Hodgkinson, Wu, and Hazel³¹, building on the birefringent-thin-film formalism of Hodgkinson and Wu³². This describes a uniaxial medium with the optic axis along the column direction; it is a local, aligned-column approximation and does *not* capture the rotating optic axis, circular birefringence, or circular dichroism of a true helical (chiral) film, nor pitch-scale resonances or spatial dispersion. The quasi-static homogenisation underlying MG requires the structural period to be small compared with the wavelength; here the helical pitch $p = 150$ nm and $\lambda \approx 500$ nm give

$p/\lambda \approx 0.30$, which is not deep in the quasi-static limit, so the calculation should be read as an order-of-magnitude screening estimate of the *local* form birefringence of an aligned columnar array rather than as the full optical response of the helical architecture. As a sensor-design hypothesis-generation illustration only, column dielectric functions are taken from tungsten ($\epsilon_W = 4.41 + 19.60i$ at $\lambda \approx 500$ nm, corresponding to optical constants $n = 3.5$, $k = 2.8^{28}$) in a vacuum host ($\epsilon_h = 1$). Both the ordinary (ϵ_o , transverse polarisation) and extraordinary (ϵ_e , longitudinal, Wiener parallel limit) dielectric functions (Eqs. 5 and 6) are evaluated as complex quantities, and the effective indices are obtained as $N = \sqrt{\epsilon}$ on the passive (positive-imaginary) branch, so that both refractive index n and extinction k are retained for each polarisation. The filling fraction $f(\alpha)$ is *not* a unique material-independent quantity: the ballistic geometry yields two distinct measures (the bead-sphere solid fraction $1 - P$ and the thresholded voxel occupancy $f_{\rho>0.5}$) that differ by up to a factor of four (Section IV E). Real GLAD morphology is moreover material-dependent through surface diffusion, sticking, re-emission, homologous temperature, deposition rate, and pressure; the substitution of W (or Mo, Co) optical constants into the Cu-scale geometry is therefore an illustrative material substitution, not a prediction of a realisable W film, and any quantitative transfer would require material-specific calibration.

III. RESULTS

A. Bead-sphere void fraction and reproducibility

This subsection addresses the reproducibility half of **Q1** (the comparison against published optical void fractions, the other half of Q1, follows in Section III D). Figure 5 shows P as a function of deposition angle for all twelve main configurations. The void fraction increases monotonically from $P = 88.1\%$ at $\alpha = 60^\circ$ (clean benchmark run; Level-A primary excluded, see Section II D) to $P = 98.5\%$ at $\alpha = 89^\circ$, consistent with the expected shadowing-enhancement at high obliquity.

Three-seed reproducibility tests at $\alpha = 80^\circ$ and $\alpha = 85^\circ$ yield coefficients of variation in atom areal density of $CV_{\rho_A} = 0.19\%$ and 0.23% , respectively (Table II). The bead-sphere void fractions at both angles are identical to one decimal place across all three seeds, demonstrating that the ballistic simulator exhibits excellent stochastic reproducibility.

B. Depth-resolved void-fraction profiles

This subsection addresses the depth-variation part of **Q2**. Figure 6 shows the layer-wise void fraction $1 - \bar{\rho}_z(z)$ (mean over the 50×50 XY grid at each Z-slice) for all eleven valid configurations.

Three features are evident:

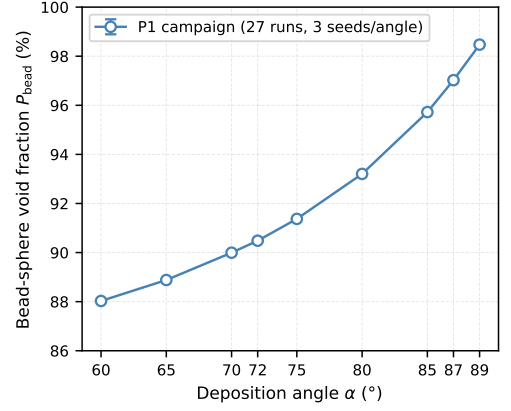


FIG. 5. Steeper deposition angles produce systematically more open (porous) films, and this trend is highly reproducible across independent random seeds. Bead-sphere void fraction P (Eq. 1), mean $\pm$ standard deviation across three independent random seeds, as a function of deposition angle for the authoritative P1 campaign (27 runs: nine angles $\alpha \in \{60, 65, 70, 72, 75, 80, 85, 87, 89\}^\circ$, three seeds each; $h = 300$ nm, pitch = 150 nm). The void fraction increases monotonically with α , reaching 98.5% at $\alpha = 89^\circ$; run-to-run reproducibility at fixed α is discussed quantitatively in Table II.

TABLE II. Three-seed reproducibility at $\alpha = 80^\circ$ and $\alpha = 85^\circ$. $\bar{P}$: mean bead-sphere void fraction; σ_P : standard deviation; CV_{ρ_A} : coefficient of variation of atom areal density.

α ($^\circ$)	n seeds	$\bar{P}$ (%)	σ_P (pp)	$\bar{\rho}_A$ (nm^{-2})	CV_{ρ_A} (%)
80	3	93.2	< 0.1	4866	0.19
85	3	95.7	< 0.1	3063	0.23

$\bar{\rho}_A$ is the deposited-atom areal density (total atoms per unit substrate area, integrated over the film height): per-seed values 4870.9/4854.8/4871.3 nm^{-2} at 80° and 3057.2/3060.1/3070.7 nm^{-2} at 85° . All three seeds give P identical to 0.1 pp.

1. *Dense substrate layer.* All configurations show a lower void fraction near the substrate ($z < 30$ nm), reflecting the initial random-accumulation regime before columnar shadowing develops. This layer is thinner and less dense at higher α .
2. *Monotonic mid-film void fraction.* The bulk columnar region (approximately $z = 50$ nm– h_{top}) shows a void fraction that increases with α , consistent with stronger shadowing. At $\alpha = 65^\circ$ the mid-film layer void fraction is approximately 57%, whereas at $\alpha = 89^\circ$ it approaches 95%.
3. *Normalised-height convergence.* When height is normalised by film top z/h (panel b), the mid-film profiles collapse more closely. Because all films share essentially the same target height ($h \approx 300$ nm), this normalisation cannot separate a growth-kinetics origin from an absolute-height origin; a height series at fixed α (available in the lev-

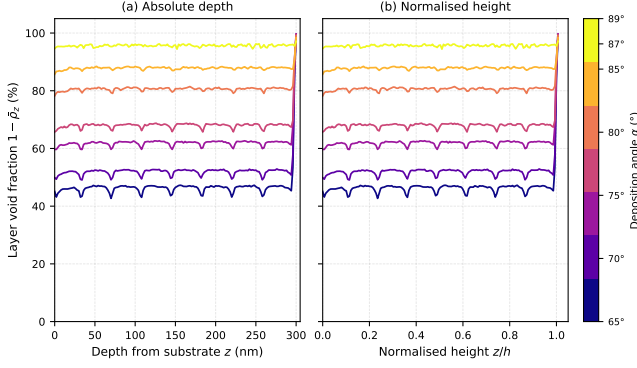


FIG. 6. Every film has a denser layer near the substrate before opening up into the angle-dependent shadowing regime that dominates the bulk of the growth. Depth-resolved void fraction $1 - \bar{\rho}_z$ as a function of (a) absolute depth from substrate (z) and (b) normalised film height (z/h), for eleven helical Cu GLAD configurations spanning $\alpha = 65^\circ$ – 89° (seed 0 for each angle; $\alpha = 60^\circ$ excluded due to a voxelisation artefact at this angle). Curves are colour-coded by α (plasma colourmap, 65° = violet, 89° = yellow). The quantity $1 - \bar{\rho}_z$ is the smeared-kernel void fraction averaged over the $100 \times 100 \text{ nm}^2$ substrate plane at each depth slice, which differs from the global bead-sphere P . All curves show a dense substrate layer ($z < 30 \text{ nm}$) transitioning to a shadowing-dominated bulk regime.

elB dataset but not analysed here) would be required to attribute the collapse to growth kinetics, and we report it only as an observed collapse. The profiles also show low-amplitude oscillations with depth; these are consistent with the helical rotation period and the finite voxel slice spacing and are not interpreted as a distinct physical growth modulation pending a dedicated sampling-convergence check.

These depth gradients are relevant to optical and sensor properties, which depend on the local void-fraction profile rather than the bulk average, although a quantitative optical use would require the convergence checks noted above.

C. Morphology latent space

This subsection addresses the low-dimensional-summary part of **Q2**. Figure 7 shows the PCA of the three non-redundant morphological descriptors extracted from the eleven valid voxel fields.

With the redundant filling fraction removed, the first two principal components account for $44.8 + 30.2 = 75.0\%$ of the descriptor variance. PC1 is now a balanced blend of all three descriptors (loadings: openness 0.65, anisotropy 0.54, width-gradient -0.53) rather than being dominated by the density measures, and PC2 is carried mainly by anisotropy and the width-gradient (-0.69 and -0.72 , openness ≈ 0). The points show a broad

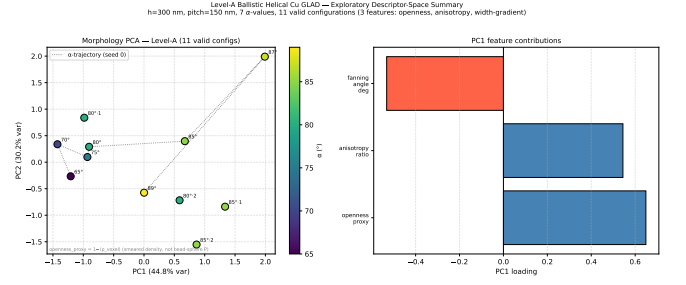


FIG. 7. Deposition angle is the dominant organising variable in this three-descriptor morphology space, but repeated seeds at the same angle still scatter visibly, showing that only the bulk density trend, not the full morphology, is tightly reproducible. (a) Principal component analysis (PCA) of three non-redundant morphological descriptors (openness $1 - \langle \rho \rangle$, anisotropy ratio, apparent width-gradient proxy) for the eleven valid ballistic helical Cu GLAD voxel fields ($\alpha = 60^\circ$ excluded due to a voxelisation artefact; the mean filling fraction is omitted as it is exactly complementary to openness). Points are colour-coded by deposition angle α ; the dotted line connects the seed-0 points in order of increasing α , and the three seeds at $80^\circ/85^\circ$ appear as separate points to show the seed-to-seed scatter. The first two principal components capture $44.8 + 30.2 = 75\%$ of the descriptor variance in this exploratory three-feature summary (with three features the per-component null expectation is 33%, so PC1 is only moderately dominant). (b) PC1 feature loadings: all three descriptors contribute comparably (openness 0.65, anisotropy 0.54, width-gradient -0.53), so PC1 is a blend rather than a single-descriptor axis.

shift toward larger PC1 as α increases, but the trajectory is not a clean monotonic locus: the three seeds at 80° and 85° are visibly scattered in the PC1–PC2 plane. This is consistent with the reproducibility analysis of Section III A, where the bulk density is reproducible to sub-percent level but the higher-order descriptors (anisotropy, width gradient) carry more run-to-run variability. Within this fixed-height, fixed-pitch sweep, deposition angle is the dominant *controlled* source of descriptor variation, but—given only eleven observations and three features, several of them repeated seeds—we read this as an exploratory summary rather than evidence of a smooth low-dimensional morphology manifold.

D. Comparison with literature void fractions

This subsection addresses the second half of **Q1**: the quantitative comparison against published optical void fractions. Figure 8 compares the ballistic bead-sphere void-fraction curve with published optical void fractions for GLAD metal films at $\alpha = 85^\circ$ from Fischer & Schubert (2013).

At $\alpha = 85^\circ$, the ballistic simulation yields $P = 95.7\%$, while the optical void fractions of GLAD metal films (Co: $75.38 \pm 0.01\%$; Ti: $78.30 \pm 0.04\%$; NiFeMo: $75.0 \pm 0.1\%$) cluster around a median of $\approx 75\%$, giving a systematic

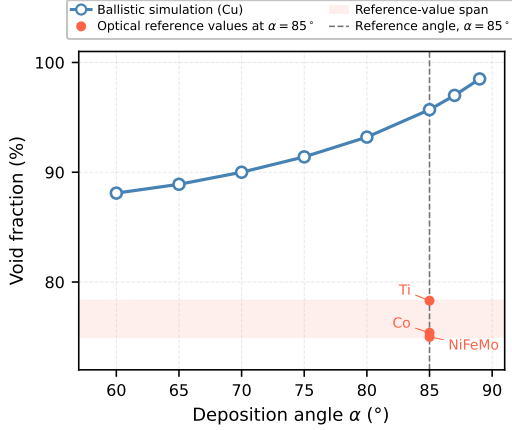


FIG. 8. The purely geometric ballistic model is systematically more open than real deposited GLAD metal films, confirming that surface diffusion and measurement-convention differences (not captured by this model) matter quantitatively. Comparison of bead-sphere void fraction P (ballistic Cu GLAD simulation, blue curve) with published optical void fractions for GLAD metal films (Co: 75.4%; Ti: 78.3%; NiFeMo: 75.0%) at $\alpha = 85^\circ$ measured by the anisotropic Bruggeman effective-medium approximation (ABEMA method; Fischer & Schubert 2013¹⁰). Red points with error bars: individual published values; shaded band: literature range. The ballistic simulation overestimates the published median by +20.7 percentage points at $\alpha = 85^\circ$. The three contributing sources of mismatch (absence of surface diffusion, measurement-convention difference, and material difference) are discussed in Section VD.

offset of $\Delta P = +20.7$ pp.

This offset is expected for three reasons:

1. *Absence of surface diffusion.* Ballistic models produce more open structures than real deposited films, where adatom mobility drives local densification after impact³³; oblique-angle-deposited films with active adatom mobility show measurably lower porosity and altered deposition-rate scaling than a purely geometric shadowing prediction would give³⁴, and this densification is known to broaden individual GLAD Cu nanorods anisotropically relative to the ballistic-limit column shape³⁵.
2. *Measurement-convention mismatch.* The bead-sphere formula counts volume occupied by rigid spheres, while optical ABEMA infers void fraction from effective-medium theory applied to ellipsometric data; the two conventions do not coincide exactly even for the same physical pore structure.
3. *Material mismatch.* The literature values are for Co, Ti, and NiFeMo; Cu has a different diffusion length and sticking coefficient, so a direct quantitative comparison would require Cu-specific experiments (structure-conduction property relations for a related Cu-based nanostructured system are

characterised in El Beainou³⁶).

Taken together, these three mismatches preclude a direct quantitative comparison. The +20.7 pp difference observed at $\alpha = 85^\circ$ should be read as an *indicative upper-bound offset* rather than a calibrated correction: Cu-specific GLAD void-fraction measurements would be required to establish a material- and method-consistent calibration reference.

A Cu-specific angle-dependent porosity series is in fact available for a related material system: Ben Nacer *et al.*³⁷ report an optically-inferred (Bruggeman effective-medium) porosity for GLAD Cu_2O thin films of 23%, 28%, 33%, and 37% at $\alpha = 0^\circ, 60^\circ, 75^\circ$, and 85° respectively, likewise increasing monotonically with α . These numbers are an order of magnitude smaller than, and not on the same scale as, the bead-sphere void fraction reported here (88.1–98.5% over $\alpha = 60\text{--}89^\circ$): the two are different physical quantities by construction, not two measurements of the same one. The present simulation reports an idealised geometric packing fraction of non-diffusing, non-reacting hard spheres with no material chemistry, whereas the Ben Nacer *et al.* value is an effective-medium fit to ellipsometric data from a real, post-deposition-annealed, oxidised (Cu_2O) film subject to surface diffusion, oxidation, and densification processes entirely absent from the model. No quantitative agreement in magnitude is expected or claimed between them, for the same reasons detailed for the Fischer & Schubert comparison above. The only comparison that is scientifically meaningful between these two datasets is that both quantities increase monotonically with deposition angle over an overlapping angular range: this is a qualitative trend-level agreement, not a quantitative validation of the simulation against experiment. We therefore state only that the simulated void-fraction-versus-angle trend is qualitatively consistent with the Ben Nacer *et al.* Cu_2O porosity-versus-angle trend, and make no claim that the simulation validates, confirms, matches, or reproduces the experimental porosity values themselves.

IV. STATISTICAL MORPHOLOGY VIA NEURAL IMPLICIT FIELD

The morphological descriptors of Section IIIC characterise the *bulk* structure of each film. We now ask whether the *spatial* structure of the voxel density field can be modelled by a neural implicit field (NIF)—a compact multi-layer perceptron mapping continuous coordinates to a scalar field—and, where direct voxel prediction fails, what statistical quantity can instead be learned and interpolated across deposition angles (**Q3**).

A. Voxel-level occupancy: realization-specific maps are not transferable

In any GLAD experiment or simulation, two runs deposited at the same angle α under identical macroscopic conditions produce the same statistical properties (void fraction, optical anisotropy) but different individual column arrangements in SEM images: column nucleation sites are determined by the random initial conditions of each run. The ballistic simulator exhibits the same behaviour: column positions are independently seeded by the random-number generator in every run. A model trained to predict the *absolute positions* of individual columns across independent runs is therefore attempting to reproduce a realization-specific arrangement that is re-randomised every run—analogous to predicting the exact positions of snowflakes in two separate snowfalls at the same temperature and humidity, even though the snowflake size *distribution* is reproducible. We test how far this limits a coordinate-conditioned model with a corotating-frame oracle experiment.

A coordinate-conditioned neural implicit field (NIF) was trained to predict per-voxel binary occupancy $I(\mathbf{r}; \alpha) \in \{0, 1\}$ from spatial coordinates (x, y, z) and deposition angle α , using a random Fourier-feature encoder and angle conditioning (FiLM, feature-wise linear modulation) on α . Seven configurations ($\alpha \in \{65, 70, 75, 80, 85, 87, 89\}^\circ$) were trained independently, and a multi-angle leave-one-angle-out (LOOA) variant was also evaluated.

Observation: the pooled multi-angle, multi-seed production NIF does not converge below the constant-occupancy baseline. Its training binary cross-entropy stalls near $\mathcal{L}_{\text{BCE}} = \ln 2 \approx 0.693$, the value of a maximally uninformative predictor that outputs a constant probability of 0.5; for the pooled multi-angle data (overall occupancy well below one half) this is no better than—and in fact worse than—even a constant predictor set to the bulk filling fraction, i.e. the network learns nothing useful. This plateau on its own is *not* evidence that the field is unrepresentable: it can equally arise from optimisation effects (here, early stopping before convergence and contradictory occupancy labels where the three independent seeds at 80° and 85° are pooled), so we do not treat the production training loss as a physical result. To separate representational capacity from these confounds we ran a controlled single-realization overfit (train = test) on one deterministic field ($\alpha = 70^\circ$, balanced occupancy $f = 0.50$): the same Fourier-feature coordinate network drives the BCE from the $\ln 2$ baseline to ≈ 0 and reaches an intersection-over-union of 1.00 on a 50×50 transverse slice and 0.997 on a 24^3 sub-volume, far above the trivial all-positive baseline ($\text{IoU} = f$). The tested deterministic slice and sub-volume are therefore representable to near-perfect accuracy by the coordinate network; the production $\ln 2$ plateau is consistent with optimisation limitations (early stopping) and pooled-realization label conflicts rather than representational impossibility, although

the present tests do not isolate its exact cause. The scientifically meaningful question is thus not whether one field can be fitted—the tested cases can—but whether an *absolute* occupancy map learned from one realization transfers to an independently nucleated run. To probe this we use a corotating-frame *oracle* that is not subject to optimisation failure: it measures whether the best fixed mean-template baseline—the empirical mean derotated occupancy map under the tested corotating protocol—carries any spatial information beyond the bulk filling fraction. For each α , the mean derotated occupancy map $M_{\text{train}}(x_{\text{rot}}, y_{\text{rot}})$ is accumulated over training slices $z \in [0, 121]$ (nearest-neighbour derotation by the local helix phase angle $\phi(k) = 2\pi k/p_{\text{vox}}$). We quantify spatial prediction success by the column-map overlap fraction (intersection-over-union, IoU): the fraction of voxels correctly identified as column material by both model and reference, divided by the union of all voxels identified as column by either. The reference baseline is the all-positive segmentation (every voxel labelled column material), whose intersection-over-union with the true occupancy equals the filling fraction f exactly by construction ($\text{IoU} = |\text{truth} \cap \text{all}| / |\text{truth} \cup \text{all}| = N_{\text{true}}/N_{\text{total}} = f$). The oracle IoU gain of the thresholded mean-template map above this baseline is:

$$\Delta = \text{IoU}_{\text{best}}(M_{\text{train}}) - f = 0.000 \quad \forall \alpha, \quad (2)$$

even when evaluated *on the training region itself* (Table III). This result holds despite peak-to-mean ratios of M_{train} ranging from 1.25 ($\alpha = 65^\circ$) to 6.57 ($\alpha = 89^\circ$). Figure 9 shows M_{train} directly for these two extremes: at $\alpha = 65^\circ$ the map is a low-contrast, spatially uncorrelated speckle field around the bulk filling fraction; at $\alpha = 89^\circ$ the higher peak-to-mean ratio reflects only a lower, more sharply-peaked overall occupancy, not an emergent coherent column template. Visual inspection is consistent with the zero oracle gain: in neither case does the mean derotated map contain a spatial pattern that a fixed-template predictor could exploit.

Empirically, then, the time-averaged derotated occupancy map carries no fixed spatial template that improves on the bulk filling fraction for *this* simulator’s pooled stochastic-nucleation protocol: the individual column positions are set independently in each run, so a coordinate-conditioned predictor of the form $f(x, y, z, \alpha) \rightarrow I$ has no realization-invariant target to recover beyond the uniform-occupancy baseline. We emphasise that this is an empirical finding for the present simulator and the present corotating protocol, *not* a general proof that no model can ever fit a voxel field: a single realised field is deterministic once generated and could in principle be memorised, but such an absolute fixed-template fit would not transfer to an independently nucleated run. The result bounds only *absolute, fixed spatial templates*; it does not rule out probabilistic or generative models, latent-seed-conditioned fields, distributional prediction, or equivariant architectures that target the statistics rather than the absolute positions. The defensi-

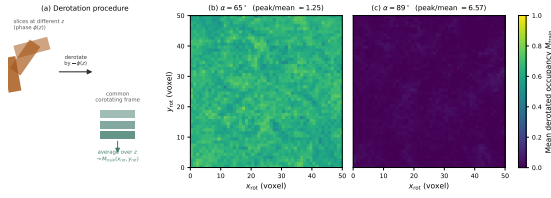


FIG. 9. The corotating-frame template itself is visually uninformative in both limits, which is why no fixed-template predictor can beat the bulk filling fraction (Table III). (a) Schematic of the derotation procedure (conceptual diagram): each z -slice is derotated by the local helix phase $-\phi(z)$ into a common corotating frame, then averaged over z to give $M_{\text{train}}(x_{\text{rot}}, y_{\text{rot}})$. (b),(c) The resulting mean derotated occupancy map for the two extreme peak-to-mean angles, $\alpha = 65^\circ$ (peak/mean = 1.25) and $\alpha = 89^\circ$ (peak/mean = 6.57), accumulated over training slices $z \in [0, 121]$. Neither map shows a coherent, learnable spatial template: the 65° map is uncorrelated speckle around a high bulk occupancy, and the 89° map’s higher peak-to-mean ratio reflects a lower overall occupancy rather than emergent structure.

ble conclusion is that absolute fixed spatial templates do not transfer between the tested independently nucleated realizations, whereas selected ensemble statistics (Section IV B) are far more reproducible. In experimental terms, this is equivalent to finding that, after derotation into the corotating frame, the column positions in one GLAD film provide no template that predicts the column positions of a nominally identical film grown in a separate run.

TABLE III. Corotating-frame oracle. M_{train} : mean derotated occupancy from training slices $z \in [0, 121]$. $\Delta_{\text{in}}/\Delta_{\text{ho}}$: IoU gain above the all-positive baseline (IoU = f) on training / holdout ($z \in [121, 151]$) regions. All gains are ≤ 0 , showing that the best fixed mean-template baseline (the mean derotated map, thresholded) does not exceed the all-positive baseline for this simulator’s pooled stochastic-nucleation protocol. $f_{\text{train}}/f_{\text{hold}}$ here use the $z \in [0, 121]/[121, 151]$ voxel-index convention of this table only; the companion study³⁸ reports occupied fractions over a different, full-height z -range at the same α , so its numeric values (e.g. 0.177 vs. this table’s 0.126 at $\alpha = 85^\circ$) are not directly comparable without accounting for that convention difference. An IoU gain of zero means that the best fixed spatial template does no better than the trivial all-positive segmentation—equivalent to finding that individual column positions carry no recoverable fixed spatial pattern in the derotated image of this film. This bounds fixed templates only, not probabilistic or equivariant predictors of the column statistics.

α ($^\circ$)	f_{train}	f_{hold}	Δ_{in}	Δ_{ho}	peak/mean
65	0.635	0.585	0.000	0.000	1.25
70	0.502	0.467	0.000	0.000	1.37
75	0.292	0.282	0.000	0.000	1.58
80	0.233	0.230	0.000	0.000	1.65
85	0.126	0.130	0.000	0.000	2.49
87	0.076	0.075	-0.0001	-0.0006	2.63
89	0.012	0.007	-0.0008	-0.0024	6.57

B. Two-point pair correlation and CorrNIF interpolation

Since individual voxel occupancies are stochastic, we shift the target to the *ensemble-level* statistical morphology: the normalised two-point pair correlation $h_g(r; \alpha)$. This real-space descriptor is related, through Fourier transformation, to the structure-factor $S(\mathbf{q})$ accessible by small-angle or grazing-incidence X-ray scattering (SAXS/GISAXS) on nanostructured films³⁹; the connection is subject to the form factor of the columns, their orientation distribution, and instrumental resolution, so $h_g(r)$ is not itself a directly measured pair-distribution function but a quantity Fourier-conjugate to the measurable scattering intensity, in the same sense as the pair-distribution function of a simple liquid is Fourier-conjugate to its static structure factor⁴⁰. It quantifies how the correlation of column material decays with radial distance r from a reference point in the transverse plane. Unlike individual column positions (Section IV A), $h_g(r; \alpha)$ is reproducible at the descriptor level across independent simulation runs at a given α , and varies smoothly with deposition angle. We verify this reproducibility directly using the three independent seeds available at $\alpha = 80^\circ$ and 85° , computing $h_g(r)$ separately for each seed. To avoid the optimistic bias of comparing each seed only against a mean that contains it, we report *independent* estimators: the mean pairwise (seed-to-seed) root-mean-square deviation is 0.0077 at 80° and 0.0090 at 85° , and the leave-one-seed-out deviation (each seed versus the mean of the other two) is 0.0066 and 0.0078, respectively (maximum pointwise standard deviation 0.010 at 80° , 0.007 at 85°). The self-inclusive deviation used in earlier drafts (0.002–0.007) understates the true variability. The CORRINF leave-one-angle-out error at these angles (RMSE $\approx$ 0.011, Table IV) is therefore comparable to, and about 1.2–1.7 $\times$ above, the independent seed-to-seed scatter: the curve is a genuinely reproducible ensemble descriptor, and the emulator reproduces it to within a small multiple of the simulator’s own run-to-run variability. The single-seed configurations at the remaining angles are used as-is; extending the three-seed test to the full angular range is identified as future work.

Definition. After bilinear derotation the corotating-frame field is a *continuous* occupancy field $J(\mathbf{r}) \in [0, 1]$ with mean $f = \langle J \rangle$ (equal to the binary filling fraction up to the interpolation error). We use the normalised covariance

$$C(r) = \langle J(\mathbf{x}) J(\mathbf{x} + r) \rangle, \quad (3)$$

$$h_g(r) = \frac{C(r) - f^2}{C(0) - f^2}, \quad h_g(0) = 1, \quad (4)$$

where $C(0) = \langle J^2 \rangle$ is read empirically from the zero-lag pixel. This continuous-covariance form (a standard normalised covariance) guarantees $h_g(0) = 1$ *without* requiring the binary identity $\langle J^2 \rangle = \langle J \rangle$, which the interpolated field does not satisfy: empirically $C(0)/f^2$

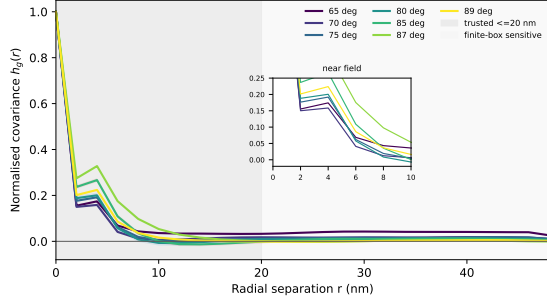


FIG. 10. The column-arrangement statistics change smoothly and systematically with deposition angle, unlike the individual column positions themselves, making this curve a promising reproducible target for interpolation across angles. Normalised two-point pair correlation $h_g(r; \alpha)$ in the corotating frame, for all seven deposition angles ($\alpha = 65^\circ$ – 89° , viridis colour scale; dense to sparse). This real-space descriptor is Fourier-conjugate to the structure factor $S(\mathbf{q})$ probed by SAXS/GISAXS: it quantifies how the correlation of column material decays with radial distance r , normalised to unity at $r = 0$. Curves differ systematically in amplitude and tail decay across α , providing the smooth signal for pair-correlation emulation; shaded uncertainty bands are shown where replicated corrected curves are available. All curves satisfy $h_g(0) = 1$ by construction and take small negative values at finite r where the arrangement is anticorrelated; the long-range tail is a finite-box ($r \leq L/2$) estimate, not a measured $h_g(\infty) = 0$.

ranges from 1.35 (65°) to 59 (89°) and differs by 15–25% from the binary value $1/f$ at every angle, confirming that the normalisation must use the measured variance rather than $1/f$. Contrary to a naive expectation, $h_g(r)$ is *not* confined to $[0, 1]$: $C(r)$ may fall below f^2 where the arrangement is anticorrelated, so $h_g(r)$ takes small negative values at finite separation (down to ≈ -0.013 at 80 – 89° here). For an infinite statistically homogeneous system the correlation is expected to approach zero at large separation; the present periodic domain only samples $r \leq L/2$, so the long-range value is not a measured $h_g(\infty) = 0$ but a finite-box estimate. Equivalently, $h_g(r) = [C(r) - f^2]/[C(0) - f^2]$ is the connected correlation normalised by its own zero-lag value; its radial *shape* remains numerically sensitive as $f \rightarrow 0$ ($\alpha \rightarrow 90^\circ$), where both $C(r) - f^2$ and $C(0) - f^2$ become small. By construction $h_g(0) = 1$ for all f , so $h_g(0)$ encodes no filling-fraction information; the filling fraction is supplied separately from the binary field.

$h_g(r; \alpha)$ is computed in the corotating frame using bilinear derotation and the FFT-based circular autocorrelation, then radially averaged over the transverse plane (full pipeline in Section II F). Figure 10 shows the resulting curves for all seven configurations. *Derotation convention.* The curves shown, the cross-seed statistics, the leave-one-angle-out CORRNF retraining, and the corrected 72° baselines (Section IV C) all use the periodic-wrap derotation and the continuous-covariance normal-

isation of Eqs. 3–4. Negative values of h_g are retained rather than clipped.

CORRNIF architecture (see Section II F for full specification). A compact NIF maps $(\alpha_{\text{norm}}, r_{\text{norm}}) \rightarrow h_g$, with a random Fourier encoder on r ($n_{\text{freq}} = 12$, $\sigma_r = 2^{26}$), three hidden layers (width 64, SiLU), and angle conditioning (FiLM) on α .

LOOA validation. In a seven-fold leave-one-angle-out (LOOA) experiment, CORRNF is trained on six angles and used to predict $h_g(r; \alpha_{\text{held}})$. Figure 11 shows LOOA predictions against the reference curves. Results are summarised in Table IV.

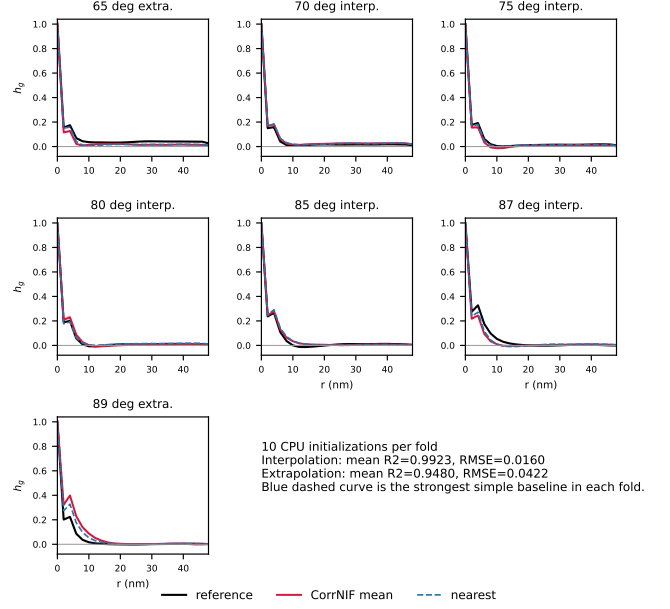


FIG. 11. The trained emulator successfully reconstructs the column-arrangement statistics at deposition angles it never saw during training, but so do much simpler interpolation methods, so its practical value lies in convenience rather than superior accuracy. Leave-one-angle-out CORRNF predictions of $h_g(r; \alpha_{\text{held}})$. In each panel, the emulator was trained on the six remaining deposition angles and had no access to the held-out angle—analogous to fitting an optical dispersion model to six incidence-angle spectra and predicting the seventh. Black solid lines: corrected reference curves computed from voxel data. Crimson solid lines and bands: mean and one-standard-deviation range of ten independently initialised CORRNF trainings. Blue dashed lines: the strongest simple baseline for the corresponding fold, evaluated against the same corrected target. Panel labels identify interpolation and extrapolation folds. The emulator interpolates the interior folds accurately but does not outperform the strongest simple baselines on accuracy.

The corrected LOOA results confirm that $h_g(r; \alpha)$ is a smoothly learnable function of α in the interior of the angle sweep. For the five interpolation folds, CORRNF reaches mean $R^2 = 0.9923$ (median 0.9954, worst case 0.9768). Crucially, this smoothness is not unique to the neural emulator: on the same corrected folds, nearest-

TABLE IV. Leave-one-alpha-out (LOOA) performance of CORRNIIF versus interpolation baselines evaluated on the *same corrected* folds: nearest-neighbour (NN, copy the training curve at the closest α), linear blend (interpolate/extrapolate between the two nearest training angles), Gaussian-process regression (GP, squared-exponential kernel, fitted per radial bin), and monotone cubic (PCHIP) interpolation along α . R^2 : coefficient of determination on the held-out curve $h_g(r; \alpha_{\text{held}})$. The two sweep-boundary folds (65° , 89°) are *extrapolation*; the five interior folds are *interpolation* and are aggregated separately. CORRNIIF entries are the mean predictions from ten independent neural initialisations. CORRNIIF does not exceed the best simple baselines on accuracy: its interior mean R^2 (0.9923) is numerically comparable to linear and PCHIP (0.9928), while the extrapolation mean drops to 0.9480. The half-width error $|\Delta d_{\text{col}}| = 2|\Delta r_{1/2}|$ is omitted because $r_{1/2} \approx 1.2 \text{ nm} \lesssim 1 \text{ vox}$ at all angles, below the 2 nm voxel resolution, so it is a sub-voxel curve-shape diagnostic rather than a physical column diameter.

$\alpha_{\text{held}} (^\circ)$	CORRNIIF	NN	Linear	GP	PCHIP
<i>Interior folds (interpolation)</i>					
70	0.9969	0.9872	0.9960	0.5602	0.9969
75	0.9954	0.9973	0.9994	0.8604	0.9987
80	0.9970	0.9989	0.9974	0.9283	0.9983
85	0.9953	0.9801	0.9932	0.9830	0.9899
87	0.9768	0.9810	0.9777	0.9852	0.9802
<i>Interior mean</i>	<i>0.9923</i>	<i>0.9889</i>	<i>0.9928</i>	<i>0.8634</i>	<i>0.9928</i>
<i>Interior median</i>	<i>0.9954</i>	<i>0.9872</i>	<i>0.9960</i>	<i>0.9283</i>	<i>0.9969</i>
<i>Boundary folds (extrapolation)</i>					
65	0.9805	0.9865	0.9865	-3.5859	0.9754
89	0.9155	0.9699	0.9699	0.8431	0.9267
<i>Boundary mean</i>	<i>0.9480</i>	<i>0.9782</i>	<i>0.9782</i>	<i>-1.3714</i>	<i>0.9511</i>
All-fold mean	0.9796	0.9858	0.9886	0.2249	0.9809

All R^2 values are coefficients of determination on the held-out curve. Interior interpolation is accurate and method-insensitive ($R^2 \approx 0.99$ for CorrNIIF, linear, and PCHIP); boundary extrapolation degrades, especially for CORRNIIF at 89° ($R^2 = 0.916$). The per-bin GP baseline is unstable for the corrected target under this small-angle training set, especially at the 65° boundary. Cubic spline interpolation (not shown) gives all-fold mean $R^2 = 0.957$.

neighbour, linear, and PCHIP baselines also reproduce the held-out curves about equally well (Table IV). We therefore do not claim that CORRNIIF interpolates *more accurately* than these simpler methods—it does not. Its advantage is qualitative: it provides a single differentiable, continuous, multi-output representation $h_g(r; \alpha)$ that can be queried and differentiated at arbitrary (α, r) and extended to additional outputs, which is convenient for downstream gradient-based optimisation but not necessary for the present interpolation task, where a simpler method would suffice. The extrapolation folds are weaker: the CORRNIIF boundary mean is $R^2 = 0.9480$, with the lowest performance at $\alpha = 89^\circ$ ($R^2 = 0.9155$), the most extreme sparse-boundary case in the sweep. This underscores that no single interpolator dominates and that the choice should be driven by the downstream use rather than by held-out accuracy alone.

C. Prospective interpolation at $\alpha = 72^\circ$

As a prospective check, we query CORRNIIF—trained on all seven existing configurations $\alpha \in \{65, 70, 75, 80, 85, 87, 89\}^\circ$ —at the deposition angle $\alpha = 72^\circ$, which is genuinely absent from the training set. This point lies 40% of the way between the nearest training neighbours 70° and 75° , making it a mid-range interpolation rather than an extrapolation. Figure 12 shows this calibration design directly.

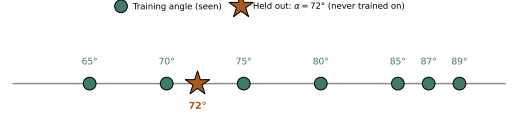


FIG. 12. The 72° test is a genuine blind interpolation, not a relabelled training point: this angle contributes nothing to CORRNIIF’s parameters before the comparison in Figure 13 is made. CORRNIIF’s training grid (green circles) versus the held-out prospective test angle (orange star, $\alpha = 72^\circ$), which is excluded from training and from all main-dataset descriptive analyses (Section II B).

Figure 13 compares the corrected CORRNIIF prediction at 72° against the dedicated reference simulation and the strongest simple baseline, a linear blend $h_{\text{lin}}(r; 72^\circ) = 0.6 h_g(r; 70^\circ) + 0.4 h_g(r; 75^\circ)$. The CORRNIIF prediction interpolates smoothly between the two neighbouring training curves but is not more accurate than this linear baseline on the corrected target.

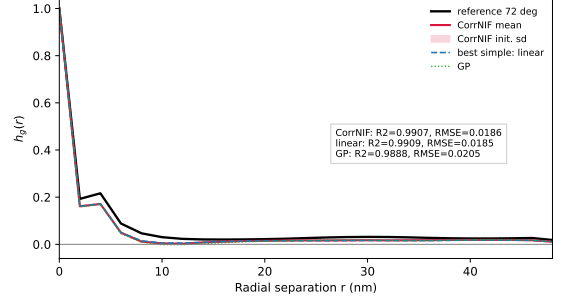


FIG. 13. A genuinely new deposition angle, never used for training and subsequently confirmed by an independent simulation, is predicted accurately by the emulator—though again no more accurately than a simple linear interpolation between the two nearest trained angles. CORRNIIF prospective prediction versus dedicated reference simulation at $\alpha = 72^\circ$ (absent from the training set). Black solid: reference simulation $h_g(r; 72^\circ)$ ($h = 300 \text{ nm}$, 68×10^6 atoms). Crimson solid and shaded band: mean and one-standard-deviation range of ten CORRNIIF initialisations ($R^2 = 0.9907$ for the mean curve). Orange dotted: linear-blend baseline ($R^2 = 0.9909$). Green dash-dotted: GP baseline ($R^2 = 0.9888$). The comparison uses the same corrected periodic-wrap, continuous-covariance target for every method.

To validate this prospective prediction, a dedicated

simulation at 72° was subsequently run under identical conditions (300 nm , 68×10^6 deposited atoms). Evaluated against this dedicated reference simulation on the corrected continuous-covariance target, CORRIF gives $R^2 = 0.9907$ (RMSE 0.0186), while the linear blend gives $R^2 = 0.9909$ (RMSE 0.0185), the best of the simple methods tested here. Nearest neighbour, PCHIP, cubic spline, and GP give $R^2 = 0.9882, 0.9902, 0.9904$, and 0.9888 , respectively. We therefore do *not* claim that CORRIF captures a non-linear α -dependence of the correlation tail that the simple interpolators cannot reproduce: for this single point they are numerically comparable. The result confirms that CORRIF generalises to a genuinely unseen interpolation point, with no demonstrated accuracy advantage over simple interpolation.

D. Demonstrating the differentiability advantage: gradient-based recovery of α

Sections IV B–IV C establish that CORRIF’s only demonstrated advantage over simple, non-differentiable baselines is qualitative: because it is a continuous, differentiable function of α , it supports gradient-based queries that nearest-neighbour, linear, and PCHIP interpolation do not. We demonstrate this advantage directly with a minimal inverse-design experiment, rather than leaving it as an unexercised claim.

Task. Treat the dedicated-simulation-confirmed pair-correlation curve at $\alpha = 72^\circ$ (Section IV C, genuinely absent from CORRIF’s training set) as an observed target, and recover the deposition angle that produced it by gradient descent through the frozen, already-trained CORRIF model—treating α itself as the only free parameter, with model weights held fixed. This mirrors a practical use case: given a desired or measured correlation shape, find the process angle that reproduces it without re-running the ballistic simulator or performing a discrete grid search.

Procedure and cross-check. Starting from five deliberately mismatched initial guesses spanning the training range ($\alpha_0 \in \{65, 70, 80, 85, 89\}^\circ$), we minimise the mean-squared error between CORRIF’s output and the target curve using Adam (300 steps, learning rate 0.05 on the normalised angle). As an independent correctness check, we also compute the loss on an exhaustive fine grid (0.005° resolution) over the full training range.

Result. All five restarts converge to the identical recovered angle, $\alpha_{\text{rec}} = 71.183^\circ$ (standard deviation across restarts $< 10^{-4}^\circ$), regardless of starting point—including restarts initialised on the opposite side of the true angle from each other (Figure 14). This matches the independent brute-force grid optimum, 71.185° , to within 0.002° , confirming that gradient descent converges to the same optimum as exhaustive search rather than to an unrelated stationary point. Each restart converges in under 4 seconds on CPU (300 Adam steps); the one-time cost of training CORRIF itself is under one minute. We report

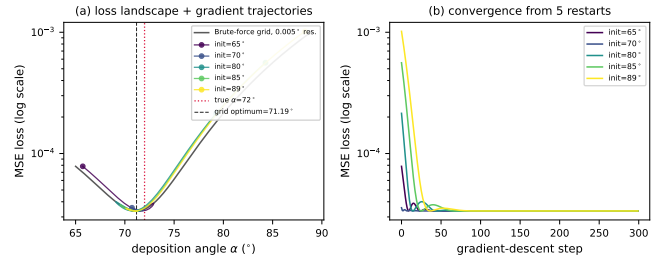


FIG. 14. Gradient descent through the frozen CORRIF model recovers the same angle from every tested starting point, and that angle matches an independent exhaustive grid search to within 0.002° —direct evidence that the differentiability claimed for CORRIF is usable, not just structurally present. (a) Brute-force loss landscape $L(\alpha)$ (grey) with the five gradient-descent trajectories overlaid, all converging to the same minimum regardless of starting point. (b) Per-restart loss-versus-iteration convergence, all five restarts reaching the same final loss within 300 steps.

this timing only to show the inverse query itself is cheap once trained, not as a claim that training plus inverse recovery is faster than a single ballistic simulation at a known angle.

We do *not* claim this recovers the true angle exactly: the recovered 71.183° differs from the actual deposition angle, $\alpha = 72^\circ$, by 0.82° . We do not claim this residual is formally comparable in magnitude to the curve-fit error reported for this same target curve in Section IV C ($R^2 = 0.9907$, RMSE = 0.0186): an angular residual (degrees) and a curve-shape RMSE (dimensionless h_g units) are different quantities, and no error-propagation relating them is derived here. Qualitatively, the 0.82° residual is small relative to the 5° spacing between the nearest training angles ($70^\circ, 75^\circ$), consistent with the inverse recovery being limited by the same finite-capacity, finite-training-grid interpolation imperfection as the forward prediction, rather than by a failure of the optimisation procedure itself, which is independently validated by the exact grid-search agreement above. The demonstration therefore substantiates the differentiability advantage as claimed: a genuine gradient-based query is usable in practice, at a precision commensurate with, and no worse than, CORRIF’s already established forward-interpolation accuracy.

E. Effective optical properties: a loss-aware Maxwell-Garnett screening illustration

This subsection addresses Q4. We compute effective optical constants from the filling fractions $f(\alpha)$ using the aligned-cylinder Maxwell-Garnett (MG) effective-medium approximation²⁷ (see Section II F for the model setup and its limitations). As a sensor-design hypothesis-generation illustration only, we take tungsten columns ($\epsilon_W = 4.41 + 19.60i$ at $\lambda \approx 500\text{ nm}$, i.e. $n = 3.5$, $k =$

2.8^{28}) in vacuum, with structural $f(\alpha)$ extracted from the Cu GLAD simulations. This screening calculation uses the $R = 0.128$ nm (Cordero covalent-radius) campaign rather than the $R = 0.1$ nm campaign used for the void-fraction, reproducibility, and PCA results elsewhere in this paper (Table I); the two radii give systematically slightly different filling fractions (void fraction ~ 0.3 – 0.5 percentage points lower at $R = 0.128$ nm), which does not change any qualitative conclusion drawn here but should not be conflated with the primary-campaign numbers reported in Section III A. Two further caveats govern the interpretation of this calculation. First, the filling fraction is definition-dependent: the bead-sphere solid fraction $1 - P$ and the thresholded voxel occupancy $f_{\rho>0.5}$ differ by up to a factor of four at low α (e.g. 0.115 vs 0.492 at 65° ; Table V), and the thresholded occupancy is itself sensitive to the Gaussian voxelisation kernel and the 0.5 threshold rather than being a physical material volume fraction. Second, the columns are strongly absorbing, so the extraordinary polarisation carries a large extinction k_e that must be reported alongside Δn . We therefore present the optical results as an illustrative, definition- and material-dependent screening calculation, not a quantitative design prediction, and report both filling-fraction conventions.

The ordinary index n_o (transverse polarisation, cylindrical inclusion geometry) is obtained from:

$$\varepsilon_o = \varepsilon_h \frac{(\varepsilon_c + \varepsilon_h) + f(\varepsilon_c - \varepsilon_h)}{(\varepsilon_c + \varepsilon_h) - f(\varepsilon_c - \varepsilon_h)}, \quad (5)$$

and the extraordinary index n_e (longitudinal, Wiener parallel limit) from:

$$\varepsilon_e = f \varepsilon_c + (1 - f) \varepsilon_h, \quad (6)$$

where ε_c is the column dielectric function and $\varepsilon_h = 1$ (vacuum host). The form birefringence is $\Delta n = |n_e - n_o|$.

Figure 15 and Table VI give the results.

TABLE V. Filling-fraction reconciliation. P : bead-sphere void fraction (Eq. 1); $1 - P$: bead-sphere solid fraction; $f_{\rho>0.5}$: thresholded voxel occupancy used in the optical calculation. The two solid-fraction measures differ by up to a factor of four and are not interchangeable; $f_{\rho>0.5}$ depends on the voxelisation kernel and threshold and is not a physical material volume fraction.

α ($^\circ$)	P (%)	$1 - P$	$f_{\rho>0.5}$	$f_{\rho>0.5}/(1 - P)$
65	88.5	0.115	0.492	4.3
70	89.5	0.105	0.409	3.9
75	90.9	0.091	0.348	3.8
80	92.7	0.073	0.197	2.7
85	95.3	0.047	0.113	2.4
87	96.6	0.034	0.070	2.0
89	98.1	0.019	0.011	0.6

For the thresholded-occupancy filling fraction, the real-index form birefringence is largest near $\alpha = 65^\circ$ ($\Delta n = 0.839$). This maximum should not be read as a usable

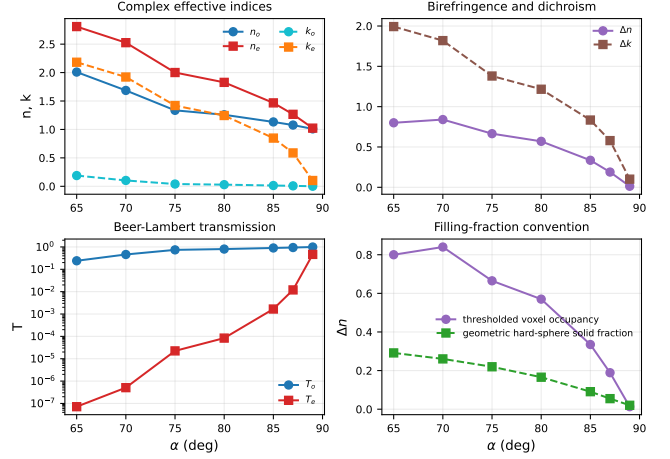


FIG. 15. A large apparent form birefringence at moderate obliquity is accompanied by near-total absorption of the extraordinary polarisation, and the apparent optimum angle is itself an artefact of how the solid filling fraction is defined—so this screening illustrates the morphology-to-optics pathway without identifying a usable polarimetric design window. Loss-aware aligned-cylinder Maxwell-Garnett optical screening of a W-column film versus α (illustrative material substitution; see Section II F). (a) Full complex effective indices for the thresholded-occupancy filling fraction: ordinary n_o and extraordinary n_e (solid), ordinary k_o and extraordinary k_e (dashed). The extraordinary extinction k_e is of order unity. (b) Form birefringence $\Delta n = |n_e - n_o|$ and dichroism $\Delta k = |k_e - k_o|$; the dichroism exceeds the birefringence across the sweep. (c) Beer-Lambert intensity transmission of the two eigenpolarisations through the 300 nm film (log scale): the extraordinary transmission T_e stays far below 1% throughout the high- Δn range, i.e. the extraordinary eigenstate is essentially fully absorbed. (d) Δn computed with the two filling-fraction conventions: the apparent maximum near 65° (thresholded occupancy $f_{\rho>0.5}$, $\Delta n = 0.839$) disappears for the geometric hard-sphere solid fraction $1 - P$, which decays monotonically. No polarimetric design window is claimed; the authoritative tabulated values are in Tables V and VI.

design optimum, for two reasons. First, it is an artefact of the filling-fraction definition: recomputing the same MG model with the geometric hard-sphere solid fraction $1 - P$ (Table V) gives a birefringence that is roughly three times smaller and *monotonically decreasing* with α ($\Delta n \approx 0.30$ at 65° falling to 0.03 at 89°), with no interior peak. The location and magnitude of the “optimum” are therefore not robust to how the solid fraction is defined. Second, and independently of the definition, the extraordinary eigenstate is almost completely absorbed across the high- Δn range: the extraordinary extinction is $k_e \approx 1.9$ at 70° , giving a Beer-Lambert intensity transmission of $T_e \sim 10^{-6}$ through the 300 nm film (Table VI). A large Δn accompanied by near-total absorption of one polarisation does not support a polarimetric device, so we do *not* claim that any angular window is suitable for guided-mode-resonance or

TABLE VI. Aligned-cylinder Maxwell-Garnett effective optical constants²⁷ for a high-index columnar film ($\varepsilon_W = 4.41 + 19.60i$, $\lambda \approx 500$ nm²⁸, vacuum host), using the thresholded voxel occupancy $f = f_{\rho>0.5}$. Both polarisations are now reported as full complex indices. n_o, k_o : ordinary index and extinction; n_e, k_e : extraordinary index and extinction; $\Delta n = |n_e - n_o|$; T_e : extraordinary intensity transmission through the 300 nm film (Beer–Lambert screening, normal incidence, interfaces neglected). The large k_e and near-zero T_e across the high- Δn range show that the extraordinary eigenstate is essentially fully absorbed; the figures are an illustrative screening calculation only (see Table V for the filling-fraction caveat).

α (°)	f	n_o	k_o	n_e	k_e	Δn	Δk	T_e
65	0.492	1.679	0.102	2.518	1.914	0.839	1.81	5×10^{-7}
70	0.409	1.522	0.070	2.320	1.728	0.798	1.66	2×10^{-6}
75	0.348	1.421	0.053	2.161	1.576	0.740	1.52	7×10^{-6}
80	0.197	1.214	0.024	1.713	1.125	0.500	1.10	2×10^{-4}
85	0.113	1.116	0.012	1.412	0.781	0.296	0.77	3×10^{-3}
87	0.070	1.070	0.007	1.241	0.551	0.171	0.54	2×10^{-2}
89	0.011	1.011	0.001	1.025	0.110	0.014	0.11	0.44

other polarimetric biosensors; establishing device suitability would require a loss-aware figure of merit and a device-level transmission or resonance calculation that are beyond the present effective-medium screening. With these caveats, the qualitative angular trend—stronger form birefringence at moderate obliquity, decaying as $f \rightarrow 0$ toward 90° —is consistent with angle-dependent optical anisotropy in GLAD copper-oxide films. In Cu_2O zigzag GLAD films grown by thermal vacuum evaporation at oblique incidence and oxidised by post-deposition annealing, a birefringence maximum of $\Delta n = 0.106$ at a deposition angle of $\gamma = 60^\circ$ (using the source studies’ own symbol γ for the same deposition-angle quantity denoted α throughout this paper; optical constants $n = 2.42$ – 2.62) has been reported¹¹. A subsequent CuO_x evaporation series reported the same angular trend: $\Delta n = 0.20$ at $\gamma = 60^\circ$ declining to $\Delta n = 0.04$ at $\gamma = 85^\circ$ for the Cu_2O phase⁴¹. These two studies share authorship and an experimental lineage, so they are corroborating rather than fully independent; they indicate that GLAD optical anisotropy in this oxide system is largest near 60° – 70° and decays at higher obliquity, consistent with the present trend. A separate deposition-angle series from an earlier study by an overlapping author group (fixed-angle, stationary-substrate GLAD, $\gamma = 0^\circ$ – 80° in five steps) reports the complementary observable of Tauc optical band gap for the Cu_2O phase within an as-deposited $\text{Cu} \rightarrow \text{Cu}_2\text{O}/\text{CuO}_x$ composite film, rising from $E_g = 2.16$ eV at $\gamma = 0^\circ$ to $E_g = 2.68$ eV at $\gamma = 80^\circ$ ⁴². This is a distinct optical quantity from the birefringence discussed above (band gap rather than form-anisotropy), so it does not directly corroborate the Δn trend, but it independently confirms that this oxide system’s optical response is strongly and monotonically angle-dependent across a wide obliquity range, consistent with the picture that GLAD-driven morphological anisotropy (column tilt and packing density) has a first-

order effect on the optical properties of Cu-oxide films regardless of which specific optical observable is measured. The much larger absolute magnitude obtained here for tungsten ($n = 3.5$, $k = 2.8$) reflects its far stronger dielectric contrast, but in view of the absorption it is not a directly realisable optical advantage and remains an illustrative material substitution pending direct W-GLAD ellipsometry. Tungsten, rather than the project’s own $\text{Cu}_2\text{O}/\text{CuO}_x$ optical constants (already used quantitatively above via^{11,41}), was chosen deliberately as a generic high-index, strongly-absorbing metal so that the Maxwell-Garnett screening in this subsection isolates the purely geometric form-birefringence mechanism from any particular material’s band-structure-specific optical response; the oxide-specific behaviour of $\text{Cu}_2\text{O}/\text{CuO}_x$ is instead addressed qualitatively via the direct literature comparison above.

V. DISCUSSION

A. Scope of the ballistic model

The results demonstrate that the ballistic GLAD simulator reproduces the expected *monotonic trend* of increasing void fraction with deposition angle across the full simulated angle range, but *quantitatively overestimates* the void fraction relative to published optical void fractions of other GLAD metals—a cross-material, cross-method discrepancy, not a calibrated offset. The model is most reliable for relative comparisons within the same simulation campaign (e.g., comparing $\alpha = 75^\circ$ to $\alpha = 85^\circ$) and less reliable for absolute predictions of experimentally measurable quantities without calibration. A Monte Carlo growth-dynamics study spanning multiple sputtered metals and oxides found that removing hyperthermal surface-relaxation processes from an otherwise similar model reproduces the thin, tilted, lower-density morphology characteristic of thermal evaporation⁴³, supporting the classification of the present purely ballistic, no-relaxation model as representative of the evaporation-like deposition regime rather than the densified sputtering regime; this is consistent with, but does not resolve, the absence of a calibrated surface-relaxation term noted above.

B. Reproducibility as a model quality criterion

The sub-0.25% CV in atom areal density across three independent seeds demonstrates that the stochastic element of the ballistic simulation (random azimuthal flux, Poisson-distributed arrival positions) does not introduce significant variability in the macroscopic void fraction at the simulated scale. This is consistent with the statistical-mechanics argument that large- N depositions converge to ensemble-average behaviour. We stress the

limited scope of this conclusion: for the global void fraction at 80° and 85° one realization appears sufficient at the reported precision, but this is not established for other angles or for higher-order descriptors (the pair-correlation curve, for example, retains a measurable seed-to-seed scatter, Section IV B). Extending the three-seed test across the full angular range remains future work.

C. Depth gradients: implications for property predictions

The dense substrate layer revealed in the Z-profiles (Section III B) would affect any optical model that integrates the refractive index through the film thickness. Effective-medium calculations that assume a spatially uniform void fraction (as typically used in¹⁰) would under-estimate the substrate-layer contribution. This observation motivates depth-graded effective-medium models as a more rigorous alternative.

D. Limitations and future work

This study is subject to the following limitations:

1. No surface diffusion: the model cannot reproduce experimental void fractions without calibration.
2. Cu-specific validation: no optical void-fraction sweep versus α is available in the published literature specifically for Cu GLAD films under conditions comparable to the present simulations; direct quantitative comparison between the simulation trend and a Cu experimental trend therefore remains future work.
3. Single helical pitch: all simulations use pitch = 150 nm; pitch dependence of void fraction is not explored here.
4. Voxelisation artefact at $\alpha = 60^\circ$: excluded from voxel analyses; root cause (kernel bandwidth mismatch) is flagged for repair in future pipeline updates.
5. Lateral box size is not fully converged. A supplementary finite-size study (Appendix A), carried out at reduced film heights ($h = 25\text{--}100\text{ nm}$ for computational tractability), shows the deposited areal density continues to decrease with lateral box size L with no plateau up to $L = 350\text{--}450\text{ nm}$, and that the run-to-run (seed) scatter *grows* with L rather than self-averaging away. Consequently the *absolute magnitudes* reported here (void-fraction endpoints, PCA loadings, CORR NIF R^2 , Maxwell-Garnett Δn) should be read as characteristic of the 100 nm-box, $h = 300\text{ nm}$ convention used throughout this study rather than as converged infinite-box values; a box exactly equal to the pitch

($L = 150\text{ nm}$) additionally shows a commensurability artefact and should be avoided. Crucially, the paper’s two central findings are *relational*, *not absolute*, and are therefore not sensitive to this magnitude caveat: (i) the oracle result that absolute column positions carry no realization-invariant signal is a statement about independence across seeds, which no change of box size can manufacture; and (ii) the smooth angular interpolability of the pair correlation (CORR NIF) is a statement about continuity in α at fixed convention. We nonetheless flag that the *quantitative* void-fraction values and the exact magnitude of the angular trend remain provisional pending an angle-resolved, full-height convergence study (future work).

Future work should address (i) addition of a phenomenological diffusion term to reduce the void-fraction overestimate relative to experiment, (ii) experimental validation with Cu GLAD films over the $\alpha = 70\text{--}89^\circ$ range, (iii) a lateral box-size convergence study specifically for the pair-correlation function used by the NIF/CORR NIF pipeline (Section IV B), to confirm the short-range correlation is free of finite-size artefacts – this remains open (the bulk atom-density convergence check described in Section V D above tests a different, coarser observable and does not by itself establish correlation-function convergence), and (iv) extension of the parameter sweep to the pitch dimension.

E. NIF scope: realization dependence versus reproducible ensemble statistics

Section IV establishes a practical hierarchy of what the voxel data can and cannot support. The oracle experiment (Table III) shows empirically that, for this simulator’s pooled corotating protocol, the best fixed spatial template (the mean derotated occupancy map) carries no information beyond the bulk filling fraction, so a coordinate-conditioned model of the form $f(x, y, z, \alpha) \rightarrow I$ has no realization-invariant target to recover: column positions are set independently in each run and do not transfer between independent realizations. Because the oracle measures the best fixed mean-template baseline rather than a particular trained network, this is stronger evidence than the flat training loss alone—it indicates that the absence of a recoverable absolute-position signal under a fixed template is a property of the stochastic growth protocol, not merely of one architecture or optimiser (it does not bound probabilistic or equivariant predictors of the statistics). This is not a claim that a voxel field is unrepresentable: our controlled single-realization overfit (Section IV A) shows that the coordinate network memorises a deterministic transverse slice and a sub-volume of one field essentially perfectly (IoU = 1.00 and 0.997, BCE $\rightarrow 0$). The limitation is specifically one of *transfer*—an absolute occupancy map fitted to one realization does not generalise to an independently nucleated

run, because the column positions are re-randomised between runs.

By contrast, the two-point pair correlation $h_g(r; \alpha)$ encodes the *ensemble statistics* of the column arrangement, which are approximately reproducible under the tested seeds and vary smoothly with α in the present dataset. The corrected CORR-NIF LOOA results (interior mean $R^2 = 0.9923$, extrapolation mean $R^2 = 0.9480$) show that this quantity is learnable in the interior of the sweep but materially less secure at the sparse-angle boundary. The prospective validation at $\alpha = 72^\circ$ ($R^2 = 0.9907$ against a dedicated simulation) further confirms that the NIF generalises beyond the leave-one-out setting to a genuinely unseen interpolation point, without demonstrating an accuracy advantage over linear interpolation ($R^2 = 0.9909$). The practical implication is that statistical descriptors—filling fraction, pair correlation half-width, column density—are the correct targets for data-driven GLAD modelling, not individual voxel occupancies.

For downstream optical screening, this finding has practical value. The illustrative form-birefringence estimates of Section IV E were derived entirely from the structural $f(\alpha)$ of the ballistic simulations, without additional experimental input; as discussed there, their absolute magnitude and the apparent angular optimum are not quantitatively defensible (filling-fraction ambiguity and strong extraordinary absorption), so they serve only to exercise the morphology-to-optics pathway rather than to define a usable sensor window. A dedicated companion study³⁸ investigates the voxel-occupancy learnability question in depth, including conditioning-based remedies and a multi-seed enlargement test. Once CORR-NIF is trained on a small set of simulation runs, it provides a continuous emulator for the correlation *shape* $h_g(r; \alpha)$ at any intermediate α . The filling fraction $f(\alpha)$ required for the Maxwell-Garnett step is supplied separately (by interpolating the measured $f(\alpha)$ values directly, as $h_g(0) = 1$ is independent of f and carries no filling-fraction information), enabling rapid effective-medium parameter sweeps without additional simulation cost. This positions the statistical-morphology NIF as a candidate differentiable component for future morphology-screening workflows: deposition angle can be swept analytically through the emulator rather than through repeated full simulations, reducing the cost of an angle sweep from N_α simulations to a single trained-network evaluation. We make no closed-loop or device-optimisation claim; the optical step remains illustrative and experimentally uncalibrated.

F. Effective-medium birefringence estimate in context

The Maxwell-Garnett calculation (Section IV E) is a loss-aware effective-medium screening illustration, not an experimental validation or a device design: it uses measured W optical constants²⁸ with geometry-derived fill-

ing fractions in an aligned-cylinder approximation that omits the rotating optic axis and chirality of the true helical film. Its quantitative outputs are limited by two factors established above: the form birefringence depends strongly on which solid-fraction definition is used (the interior Δn maximum vanishes for the geometric hard-sphere fraction), and the extraordinary eigenstate is almost fully absorbed ($k_e \sim 1$, $T_e \sim 10^{-6}$ through 300 nm). We therefore retain only the qualitative angular trend—larger form birefringence at moderate obliquity, decaying toward 90° —which is consistent with the Cu₂O GLAD measurements of Chaffar Akkari *et al.*^{11,41} (these two share authorship and lineage and are corroborating rather than fully independent). We make no claim about the absolute W-column magnitude or about polarimetric device suitability; genuine validation requires generalised (anisotropic) W-GLAD ellipsometry and a device-level optical model, both identified as primary future experiments. Furthermore, the transfer of filling fractions from Cu ballistic simulations to W, Mo, or Co columns assumes comparable columnar geometry; in practice, material-specific surface diffusion, sticking coefficients, and thermally activated adatom mobility will shift the true filling fractions and must be calibrated experimentally against material-specific GLAD depositions before quantitative predictions can be made.

The sensor-application chain is further grounded by direct evidence of Cu-oxide GLAD as a gas-sensing platform: Benamor *et al.*¹² demonstrated that Cu₂O GLAD films on glass exhibit surface roughness R_a increasing from 4.84 to 8.22 nm with deposition angle, directly confirming the $\alpha \rightarrow$ morphology $\rightarrow$ sensor response relation that the present morphology workflow targets. Together, the ballistic simulation (with its characterised cross-material discrepancy), the CORR-NIF morphology emulator, and the illustrative MG calculation form a simulation-only workflow for ranking helical GLAD geometries by ensemble morphology and screening their *candidate* optical behaviour, with the optical step explicitly flagged as illustrative pending material- and device-specific validation.

VI. CONCLUSION

We have characterised the scope of a bead-sphere ballistic GLAD simulator across twelve helical Cu configurations spanning $\alpha = 60$ – 89° and demonstrated that its morphological outputs can be partitioned into two distinct predictability classes. We close by answering, in order, the four questions (Q1–Q4) posed in Section I.

Answering Q1 (void-fraction trend, reproducibility, and comparison with experiment): the bead-sphere void fraction P increases monotonically from 88.1% to 98.5% over the tested angle range with sub-0.25% run-to-run variability at the 100 nm-box, $h = 300$ nm convention (Appendix A shows this reproducibility is convention-dependent, degrading at larger lateral box size), con-

firming that the bulk void fraction is well-defined and reproducible at the descriptor level within that convention. Against published optical void fractions of other GLAD metals at $\alpha = 85^\circ$, the simulator’s bead-sphere void fraction is +20.7 pp higher; this is a cross-material, cross-method discrepancy—attributable to the absence of surface diffusion, the differing measurement convention, and the different materials—not a calibrated offset, and Cu-specific measurements would be needed to establish a true calibration.

Answering Q2 (depth dependence and low-dimensional morphology summary): the depth-resolved void-fraction profiles show a denser substrate layer ($z < 30$ nm) giving way to a shadowing-dominated bulk regime whose void fraction increases with α , a gradient that matters for any optical model that integrates the refractive index through the film thickness. An exploratory principal-component summary of three non-redundant morphology descriptors captures 75% of their variance in two components, with deposition angle as the dominant controlled source of variation; the higher-order descriptors (anisotropy, width gradient) show visible seed-to-seed scatter, so we read this as an exploratory descriptor-space summary rather than evidence of a smooth low-dimensional manifold.

Answering Q3 (learnability of morphology by a neural implicit field): individual absolute column-position maps are not a stable target across independent realizations under the stochastic-nucleation protocol of this simulator. A corotating-frame oracle experiment shows empirically that the best fixed mean-template baseline gives zero IoU gain ($\Delta = 0.000$) over the all-positive baseline ($\text{IoU} = f$) at all seven tested angles, so absolute column positions carry no realization-invariant signal that a coordinate-conditioned model could recover. The normalised two-point pair correlation $h_g(r; \alpha)$ sidesteps this limitation by encoding ensemble statistics rather than individual absolute column positions, and is approximately reproducible within the tested three-seed cases at 80° and 85° (independent pairwise seed-to-seed RMSE 0.008–0.009, leave-one-seed-out 0.007–0.008). A compact parametric emulator (CORRNIF) trained on six deposition angles achieves an interior leave-one-angle-out mean $R^2 = 0.9923$ and an extrapolation mean $R^2 = 0.9480$, comparable to the strongest simple baselines on interpolation folds but weaker at the 89° boundary: the emulator does not improve interpolation *accuracy* over these simpler methods, and its value is instead the differentiable, continuous, multi-output representation it provides for downstream optimisation. A prospective test at $\alpha = 72^\circ$ —genuinely absent from the training set and subsequently confirmed by a dedicated simulation—yields $R^2 = 0.9907$ for CORRNIF; a simple linear interpolation between the neighbouring angles is marginally higher on the same corrected target ($R^2 = 0.9909$). This demonstrates that CORRNIF generalises beyond the leave-one-angle-out setting without a demonstrated accuracy advantage over simple interpolation.

Answering Q4 (optical-anisotropy implications of the morphology): an illustrative aligned-cylinder Maxwell-Garnett calculation maps the extracted filling fractions to candidate optical constants. We do not draw a quantitative design window from it: the apparent real-index birefringence maximum ($\Delta n = 0.839$ at 65° for the thresholded-occupancy fraction) disappears when the geometric hard-sphere fraction is used, and the extraordinary eigenstate is almost fully absorbed ($k_e \sim 1$, $T_e \sim 10^{-6}$ through 300 nm). The retained result is only the qualitative angular trend, which is consistent with the Cu_2O and CuO_x GLAD measurements of Chaffar Akkari *et al.*^{11,41} (a single shared experimental lineage, not independent confirmations). Together, the ballistic simulator (with its characterised cross-material discrepancy), the CORRNIF statistical-morphology emulator, and the illustrative effective-medium step form a simulation-only workflow for ranking helical GLAD geometries by ensemble morphology; quantitative optical design and material calibration are left to dedicated experiments. The magnitude of the surface-diffusion omission can be given an independent, quantitative anchor from an unrelated GLAD system: for sputter-deposited ZnO nanorods, LaForge⁴⁴ reports an adatom surface-diffusion hop time of order $\tau_h \sim 2 \times 10^3$ s and a monolayer-burial time $\tau_m \sim 20$ –350 s under comparable growth conditions—both far shorter than the deposition timescale of a single monolayer in a typical GLAD run, confirming that diffusive adatom motion is not a negligible process on the timescale of column formation even though it is absent from the present ballistic model by construction. This is offered as an order-of-magnitude cross-material check on the size of the omitted physics, not a claim that these particular timescales transfer quantitatively to Cu.

The principal identified limitations—absence of surface diffusion, restriction to Cu hard-sphere geometry, and the single helical pitch studied—define clear targets for future extension: addition of a phenomenological diffusion correction, experimental validation with Cu GLAD films, and exploration of pitch-dependent morphology effects.

ACKNOWLEDGMENTS

The authors acknowledge the institutional and scientific support of the Laboratoire de Photovoltaïque et Matériaux Semi-conducteurs (LPMS), École Nationale d’Ingénieurs de Tunis (ENIT), Université de Tunis El Manar, and the Institut des NanoSciences de Paris (INSP), Sorbonne Université, within the framework of the PHC-Utique project.

DECLARATIONS

- **Funding:** No specific financial support statement is declared for this manuscript because no verified project number, contract number, or funding-

agency wording was found in the local project records.

- **Conflict of interest:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** The simulation data and analysis code supporting the findings of this study are available from the corresponding author upon reasonable request.
- **Materials availability:** Not applicable.
- **Code availability:** The analysis, corrected pair-correlation, figure-generation, and CORRINF training code supporting the findings of this study are available from the corresponding author upon reasonable request.
- **Author contribution:** O.B.H.A. designed and performed all simulations, developed the NIF training pipeline, and wrote the manuscript. F.C.A., B.G., and N.M. supervised the work and critically revised the manuscript.

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(TSpace repository identifier not verified). Fe/FeS2 and ZnO GLAD systems; adatom surface-diffusion hop time and mono-layer burial time reported for ZnO (Ch.6).

Appendix A: Finite-size convergence study

This appendix reports, in full, the box-size (lateral extent L) convergence study summarised as a limitation in Section VD (Figure 16 shows the full dataset). For computational tractability this study was carried out at reduced film heights ($h = 25\text{--}100\text{ nm}$, rather than the main study's $h = 300\text{ nm}$) and reports the bulk atom areal/volumetric density directly (atoms/nm² or atoms/nm³), not the void-fraction descriptor used elsewhere in this paper; the two are related but not identical observables, so the numbers below should not be read on the same numerical scale as the main-text void fractions.

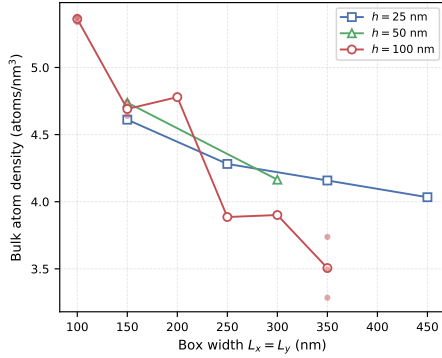


FIG. 16. Bulk atom density falls monotonically with box width at every film height tested, with no sign of a plateau at the box size used in the main study – direct visual evidence for the finite-size limitation discussed in Section VD. Bulk atom density versus box width L for all trusted runs in the finite-size convergence campaign, grouped by film height h ($\alpha = 85^\circ$, pitch = 150 nm fixed throughout). Lines connect the mean density at each L ; small dots at $h = 100$ nm, $L = 350$ nm show the individual seed-replicate values behind that mean.

A box-size sweep ($L_x = L_y \in \{75, 100, 125, 150\}$ nm at $\alpha = 85^\circ$, pitch = 150 nm fixed) shows the mean atom areal density decreases monotonically and substantially with box size (by $\approx 29\%$ from 75 nm to 150 nm), with no sign of a plateau by $L = 150$ nm (Figure 16). The 100 nm box used throughout the main study is therefore *not* at a converged, asymptotic value; the true large-box limit is likely lower still. Separately, a box size exactly equal to the pitch ($L = 150\text{ nm} = \text{pitch}$) introduces an additional, sharper commensurability artefact (erratic, non-physical atom counts at low α) distinct from the smooth finite-size trend – box sizes equal to (or simple multiples/fractions of) the pitch should be avoided in future campaigns. Attempts to extend the sweep beyond $L = 150\text{ nm}$ in this first pass were constrained by the simulator's GPU memory management rather than by physics.

A subsequent height-matched scan ($h = 25\text{ nm}$, $L \in \{150, 250, 350, 450\}$ nm) confirmed the density keeps decreasing with no plateau, and showed that a single-variable finite-size scaling collapse (density as a function of $L/(h \tan \alpha)$ alone) does not hold across different heights – at least two physical length scales (the finite-size shadow length and a substrate-proximity transient) govern the box-size dependence, so the large-box asymptote and a corrected scaling ansatz remain open questions for future work. Cross-checking the $L = 100$ vs 150 nm density change at three angles ($\alpha = 75, 85, 89^\circ$) shows this bias is *not* angle-independent (-32% , -21% , -20% respectively): the angular trend reported in the main text, while qualitatively consistent with the literature comparisons discussed there, should be read as possibly compressed or shifted in magnitude by an angle-dependent finite-size effect that a fixed box size cannot separate from the true physics.

A subsequent degeneracy-breaking campaign (three additional targeted runs: $h = 50\text{ nm}$ at $L = 150$ and 300 nm, and a second seed at $h = 100\text{ nm}$, $L = 350\text{ nm}$) extended the height-matched scan above to a third distinct film height and provided the first genuine seed-replicate at large L . Fitting a multiplicative shadow-decay $\times$ substrate-transient model, $\rho(L, h) = \rho_\infty - A(1 - e^{-h/\tau})e^{-L/\lambda}$, with a heteroscedastic weighting exponent p swept over $\{0.5, 0.75, 1.0\}$ (rather than fixed, since p is a closure-model choice, not an estimable constant, at this data volume) gives ρ_∞ point estimates of 4.28, 4.19, and 4.03 atoms/nm³ respectively, with 95% bootstrap confidence intervals that overlap across all three p ; the substrate-transient amplitude A is consistently unresolvable (pinned near zero for every p), confirming that only ρ_∞ , not the transient term, is currently informative from this data. Independently, a real-space pair-correlation length $\xi(L)$ measured directly from atom positions (not from the voxelised pipeline) is still increasing at $L = 350\text{ nm}$ across the $h = 25, 50$, and 100 nm series alike – consistent with, and independent evidence for, the absence of a finite-size plateau reported above. One run in the original height-matched scan ($h = 25\text{ nm}$, $L = 250\text{ nm}$) had originally failed an automated column-tilt validation gate; this was root-caused to a validator threshold that incorrectly ran a straight-column tilt regression on a genuinely (if only briefly) rotating deposit, producing a spurious result – the underlying atom data for that run is sound and it is retained in the trusted dataset once the validator was corrected.

A resolved but consequential finding: a third seed at $L = 350\text{ nm}$ was run specifically to place the large- L variance estimate on the same statistical footing ($n = 3$) as the $L = 100$ anchor. With this third replicate, the empirical relative seed-to-seed variance in areal density is confirmed to *grow*, not shrink, with box size: $L = 100$: 0.32% ($n = 3$); $L = 150$: 1.09% ($n = 2$); $L = 350$: 5.28% ($n = 3$) – an approximately 16 $\times$ increase in relative scatter from the smallest to the largest box, on equal ($n = 3$) statistical footing at the two endpoints,

not a two-point artefact. This is the opposite trend from the shrinking variance assumed by every p tested above (all presuppose $N_{\text{col}} \sim L^2$ independent columns self-averaging down the noise), and it is large enough to matter: refitting with a growing-variance weight $\sigma(L) \propto (L/L_{\text{ref}})^q$, with q estimated directly from these three anchors ($q \approx 2.18$) or swept over $\{1.0, 2.0, 2.5\}$, gives $\rho_\infty \approx 4.9\text{--}5.2$ atoms/nm³ – materially higher than, and non-overlapping with, the 4.0–4.3 atoms/nm³ range obtained under the originally-assumed shrinking-variance closure. We therefore report ρ_∞ as 4.9–5.2 atoms/nm³ under the empirically-supported growing-variance weighting, and flag the shrinking-variance value as based on a now-falsified assumption about how simulation noise scales with box size, retained here only for transparency about how much a single modelling choice can move this number. We do not currently have a first-principles explanation for why relative fluctuations grow with L in this simulator; the most plausible hypothesis links it to the non-saturating $\xi(L)$ reported above (a growing transverse correlation length implies the effective number of statistically-independent columns grows more slowly than L^2 , which would suppress rather than enhance self-averaging at large L), but confirming this mechanism – as opposed to merely being consistent with it – is future work. Practically, this means finite-size extrapolation in this system is noise-limited by a growing, not shrinking, per-box uncertainty at large L , so additional seeds at large L (rather than additional, more expensive large- L single runs) are the more statistically efficient way to

tighten ρ_∞ further.

Two claim tiers follow from the box-size result above and should not be conflated. *Quantitative* absolute values reported anywhere in the main text – the void-fraction endpoints (88.1%, 98.5%), the PCA variance-explained figure, the CORRIF R^2 values, and the Maxwell-Garnett birefringence magnitudes – are all computed at the 100 nm-box convention and are provisional: given the density shifts documented above, their numerical values would change at a larger, still-unconverged box size, and none should be read as an asymptotic result. *Qualitative*, monotonic angle-orderings (void fraction rising with α ; birefringence falling with α in the literature comparison of Section IV E) are the category of claim least exposed to this limitation in principle, but at present this project can only offer partial, not conclusive, support for that expectation: a box-size cross-check exists at just three angles ($\alpha = 75, 85, 89^\circ$), it is itself confounded by a simultaneously reduced active-window parameter, and within it the openness proxy shifted upward by a small, similarly-sized amount at all three angles (consistent with, but not proof of, order-preservation) while a different descriptor computed on the same box-size campaign, $\beta_{\text{sim}}(\alpha)$, was found to be non-monotonic and its box-size dependence remains unresolved. We therefore do not assert that the monotonic orderings reported here are established to be box-size-robust; they should also be treated as provisional pending an angle-resolved convergence study, and are offered as the paper’s best-supported trend-level claims rather than as finite-size-independent results.